

Spring 2015

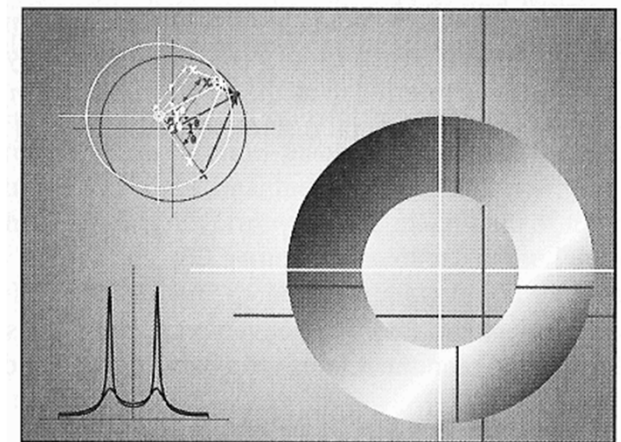
信號與系統 Signals and Systems

Chapter SS-10 The z-Transform

Feng-Li Lian

NTU-EE

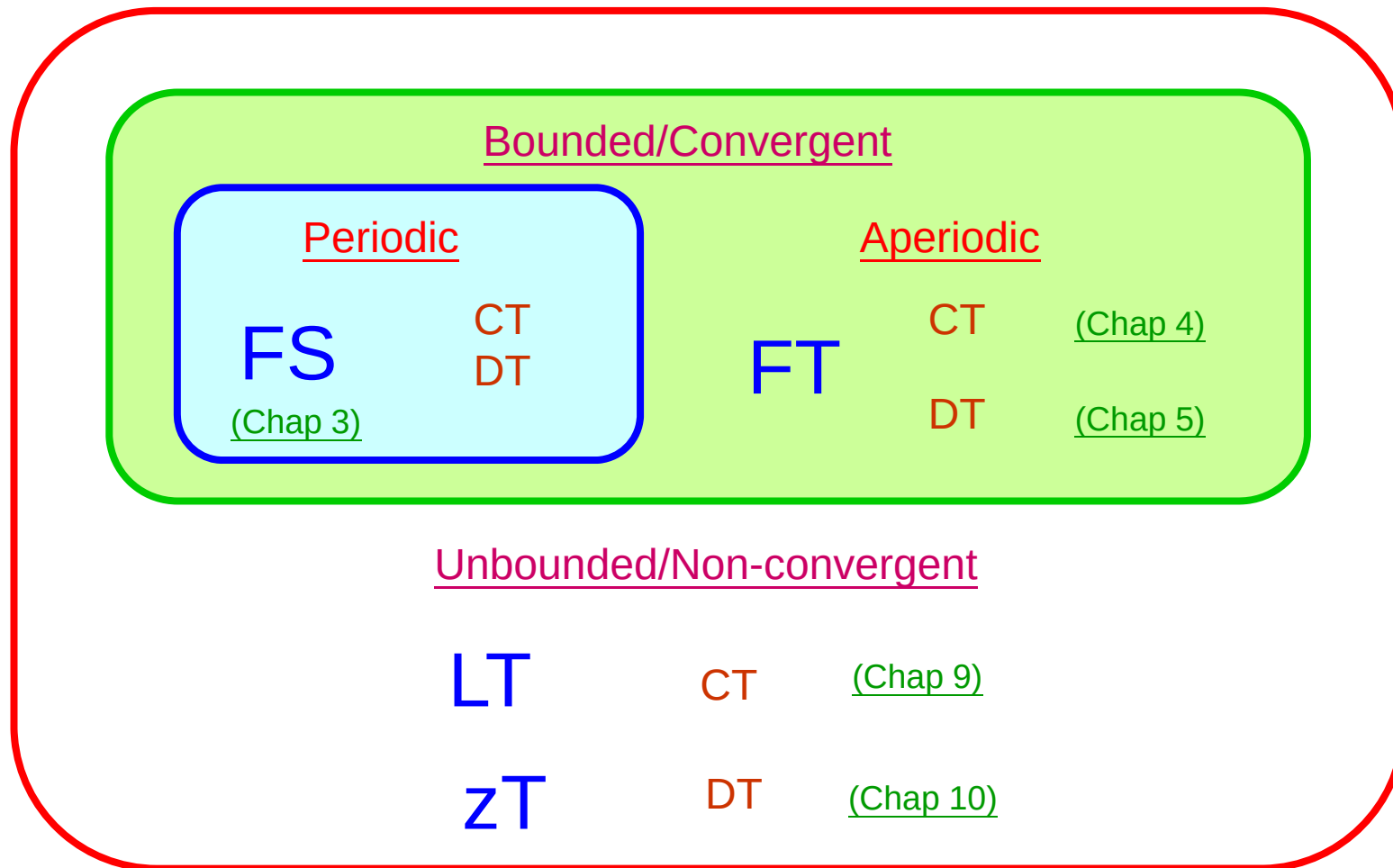
Feb15 – Jun15



Figures and images used in these lecture notes are adopted from
“Signals & Systems” by Alan V. Oppenheim and Alan S. Willsky, 1997

Introduction [\(Chap 1\)](#)

LTI & Convolution [\(Chap 2\)](#)



Time-Frequency [\(Chap 6\)](#)

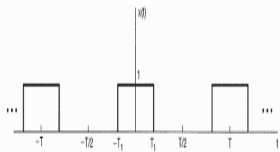
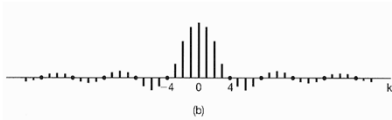
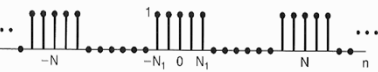
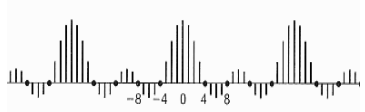
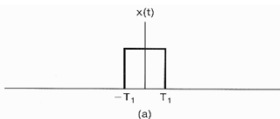
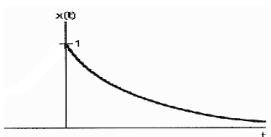
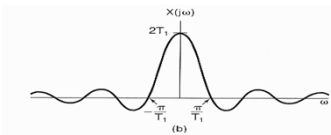
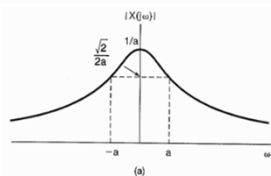
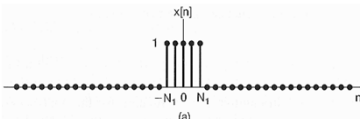
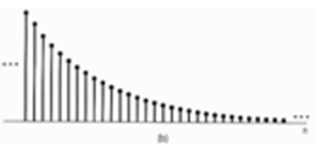
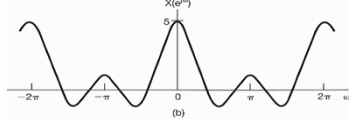
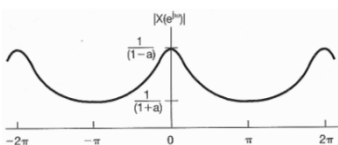
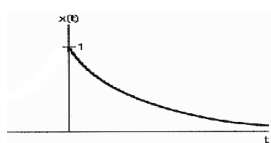
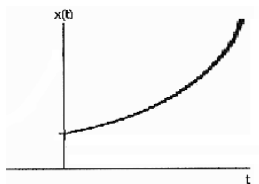
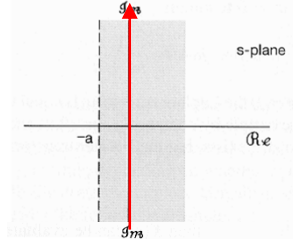
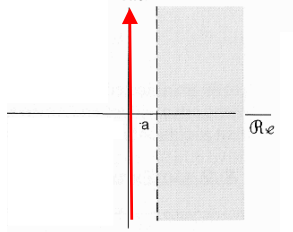
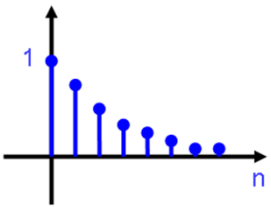
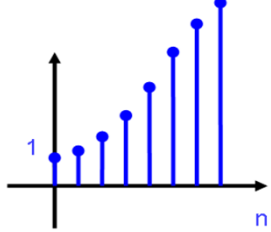
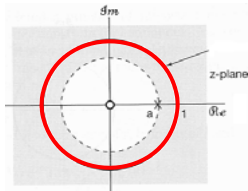
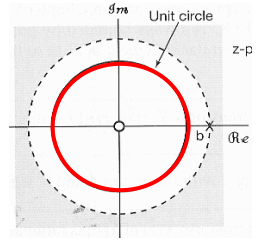
Communication [\(Chap 8\)](#)

Digital
Signal
Processing [\(dsp-8\)](#)

CT-DT [\(Chap 7\)](#)

Control [\(Chap 11\)](#)

Fourier Series, Fourier Transform, Laplace Transform, z-Transform

	CT		DT	
	time	frequency	time	frequency
FS				
FT	 	 	 	 
LT/zT	 	 	 	 

- The z -Transform
- The Region of Convergence for z -Transforms
- The Inverse z -Transform
- Geometric Evaluation of the Fourier Transform
- Properties of the z -Transform
- Some Common z -Transform Pairs
- Analysis & Characterization of LTI Systems
Using the z -Transforms
- System Function Algebra and
Block Diagram Representations
- The Unilateral z -Transform

Brief History of the z-Transform

- The z-transform was known to Laplace, and re-introduced in 1947 by W. Hurewicz as a tractable way to solve linear, constant-coefficient difference eqns.
- It was later dubbed "the z-transform" by Ragazzini and Zadeh in the sampled-data control group at Columbia University in 1952
- The name of "the z-transform"
 - The letter "z" being a sampled/digitized version of the letter "s" in Laplace transforms.
 - Another possible source is the presence of the letter "z" in the names of both Ragazzini and Zadeh who published the seminal paper.
- The modified or advanced z-transform was later developed and popularized by E. I. Jury in 1958, 1973.
- The idea contained within the z-transform is also known as the method of generating functions around 1730 when it was introduced by DeMoivre with probability theory.
- From a mathematical view the z-transform can also be viewed as a Laurent series where one views the sequence of numbers under consideration as the (Laurent) expansion of an analytic function (the z-transform).

From the Fourier Transform of DT signals $x[n]$:

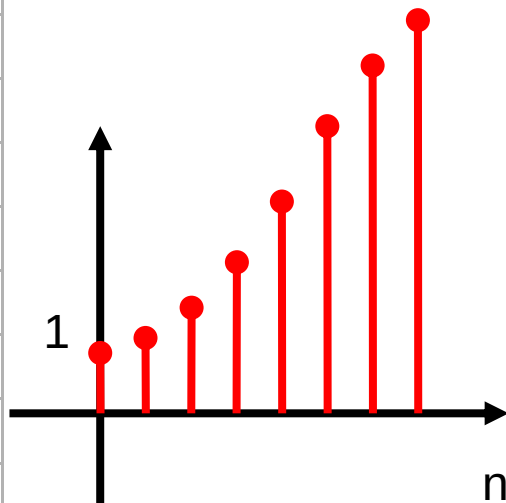
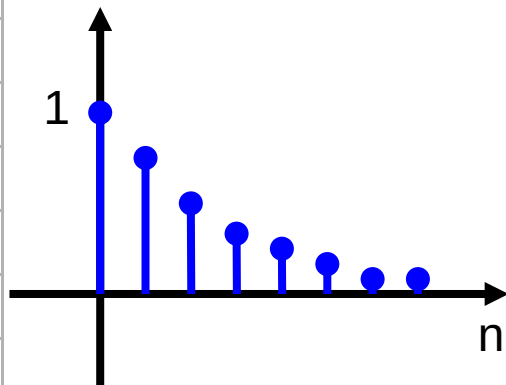
$$x[n] = a^n u[n]$$

FT

$$X(e^{j\omega}) \triangleq \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n} = \sum_{n=-\infty}^{+\infty} x[n] (e^{j\omega})^{-n}$$

$$= \sum_{n=-\infty}^{+\infty} a^n u[n] e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} (ae^{-j\omega})^n = \frac{1}{1 - ae^{-j\omega}}$$



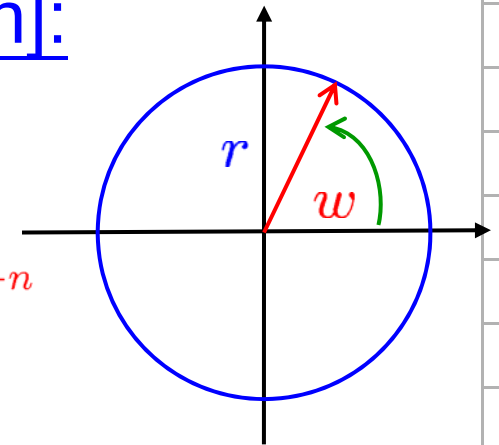
$$\rightarrow \sum_{n=-\infty}^{+\infty} (x[n] r^{-n}) (e^{j\omega})^{-n}$$

$$= \sum_{n=-\infty}^{+\infty} x[n] (r e^{j\omega})^{-n}$$

$$= \sum_{n=-\infty}^{+\infty} x[n] (z)^{-n} = X(z) \quad \text{zT}$$

■ The z-Transform of a General Signal $x[n]$:

$$\begin{array}{ll} \text{FT} & z = e^{j\omega} \\ X(e^{j\omega}) \triangleq & \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n} \end{array} \quad \begin{array}{ll} \text{zT} & z = r e^{j\omega} \\ X(z) \triangleq & \sum_{n=-\infty}^{+\infty} x[n] z^{-n} \end{array}$$



$$\begin{aligned} e^{j\omega} &= \cos(\omega) + j \sin(\omega) \\ r e^{j\omega} &= r (\cos(\omega) + j \sin(\omega)) \end{aligned}$$

$$x[n] \xleftrightarrow{\mathcal{F}} X(e^{j\omega})$$

$$x[n] \xleftrightarrow{\mathcal{Z}} X(z)$$

$$X(e^{j\omega}) = \mathcal{F} \{ x[n] \}$$

$$X(z) = \mathcal{Z} \{ x[n] \}$$

$$x[n] = \mathcal{F}^{-1} \{ X(e^{j\omega}) \}$$

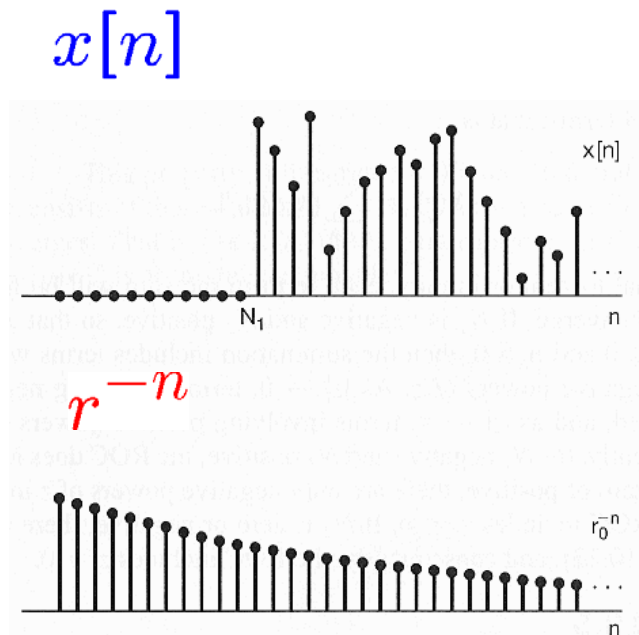
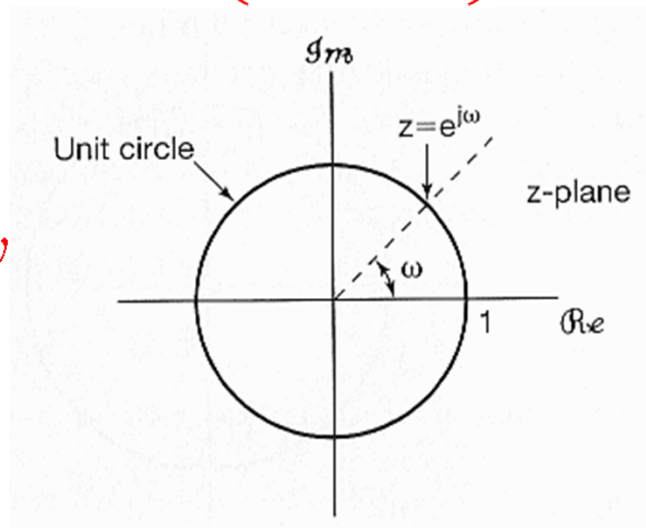
$$x[n] = \mathcal{Z}^{-1} \{ X(z) \}$$

$$X(z) \Big|_{z=e^{j\omega}} = \mathcal{Z} \{ x[n] \} \Big|_{z=e^{j\omega}} = \mathcal{F} \{ x[n] \} = X(e^{j\omega})$$

■ z-Transform & Fourier Transform:

$$\begin{aligned}
 \mathcal{Z} \left\{ x[n] \right\} \Big|_{z=re^{j\omega}} &= X(re^{j\omega}) \\
 &= \sum_{n=-\infty}^{+\infty} x[n] (re^{j\omega})^{-n} \\
 &= \sum_{n=-\infty}^{+\infty} \{ x[n] r^{-n} \} e^{-j\omega n} \\
 &= \mathcal{F} \left\{ x[n] r^{-n} \right\}
 \end{aligned}$$

$$\begin{aligned}
 z &= e^{j\omega} \\
 |z| &= 1
 \end{aligned}$$

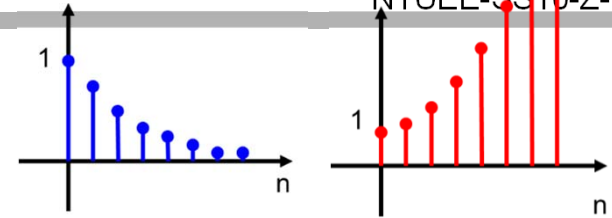


The z-Transform

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■ Example 10.1:

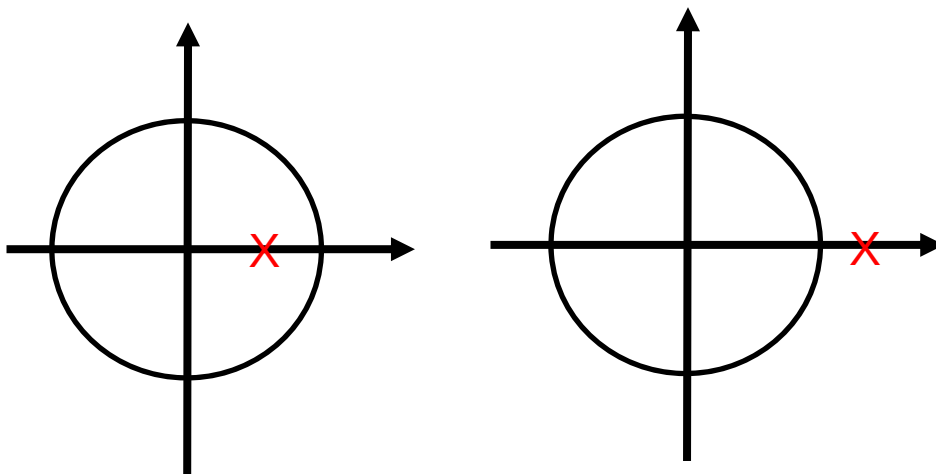
$$x[n] = a^n u[n]$$



$$\Rightarrow X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} a^n u[n] e^{-j\omega n} = \sum_{n=0}^{\infty} (ae^{-j\omega})^n = \frac{1}{1 - ae^{-j\omega}}$$

$$\Rightarrow X(z) = \sum_{n=-\infty}^{+\infty} a^n u[n] z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n \quad |az^{-1}| < 1$$

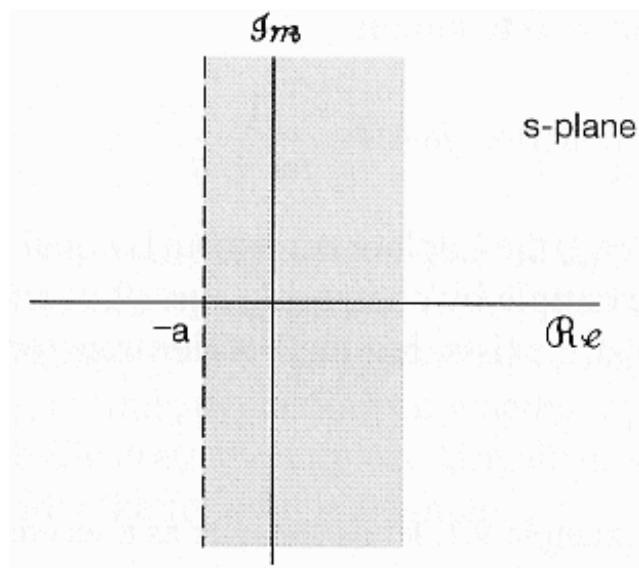
$$= \frac{1}{1 - az^{-1}} = \frac{z}{z - a}, \quad |z| > |a|$$



• For $|a| > 1$,

ROC does not
include the unit circle,

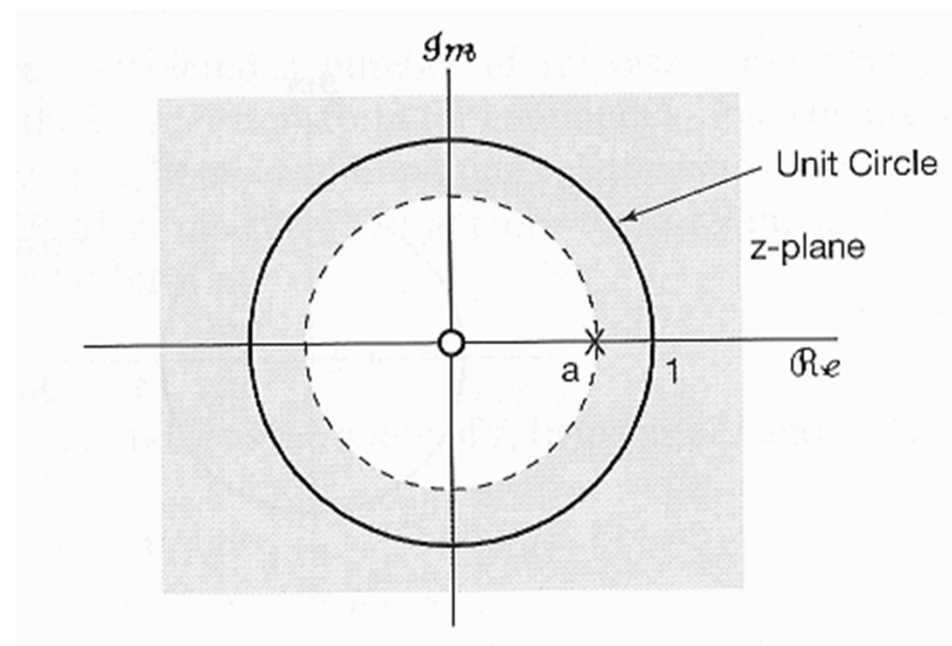
$\mathcal{F}\{a^n u[n]\}$ does not converge



$$s = \sigma + jw$$

$$e^{-at}u(t) \quad e^{-\sigma t} e^{-jw t}$$

$$e^{-st}$$



$$z = r e^{jw}$$

$$a^n u[n] \quad r^{-n} (e^{jw})^{-n}$$

$$(z)^{-n}$$

Example 10.2:

$$x[n] = -a^n u[-n-1]$$

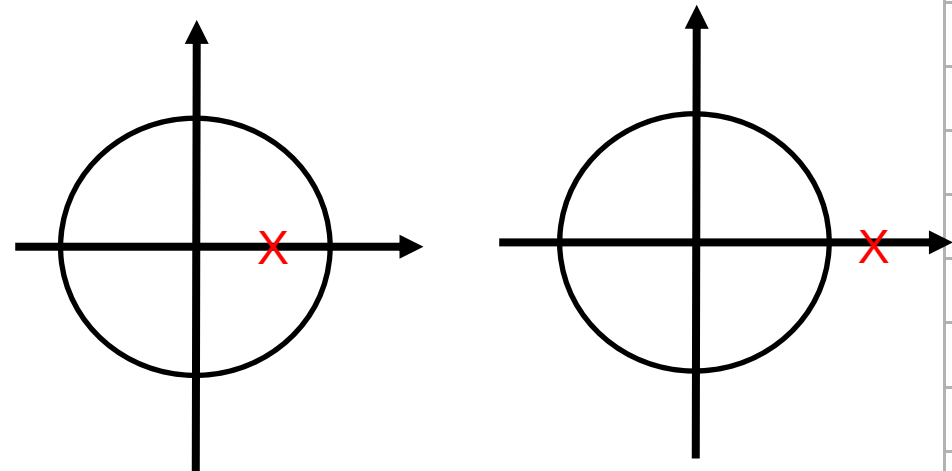
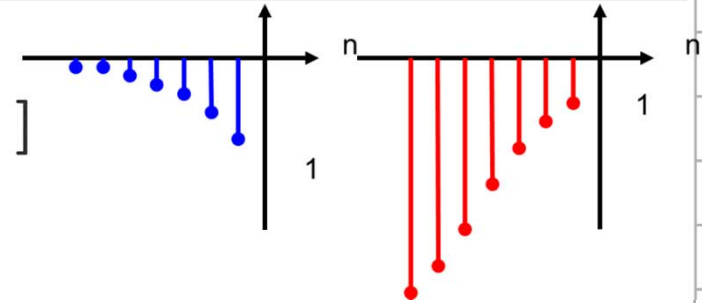
$$\Rightarrow X(z) = - \sum_{n=-\infty}^{+\infty} a^n u[-n-1] z^{-n}$$

$$= - \sum_{n=-\infty}^{-1} a^n z^{-n} = - \sum_{n=1}^{\infty} a^{-n} z^n = 1 - \sum_{n=0}^{\infty} (a^{-1} z)^n$$

$$|a^{-1} z| < 1 \quad = 1 - \frac{1}{1 - a^{-1} z}$$

$$= \frac{1}{1 - a z^{-1}} = \frac{z}{z - a}$$

$$|z| < |a|$$

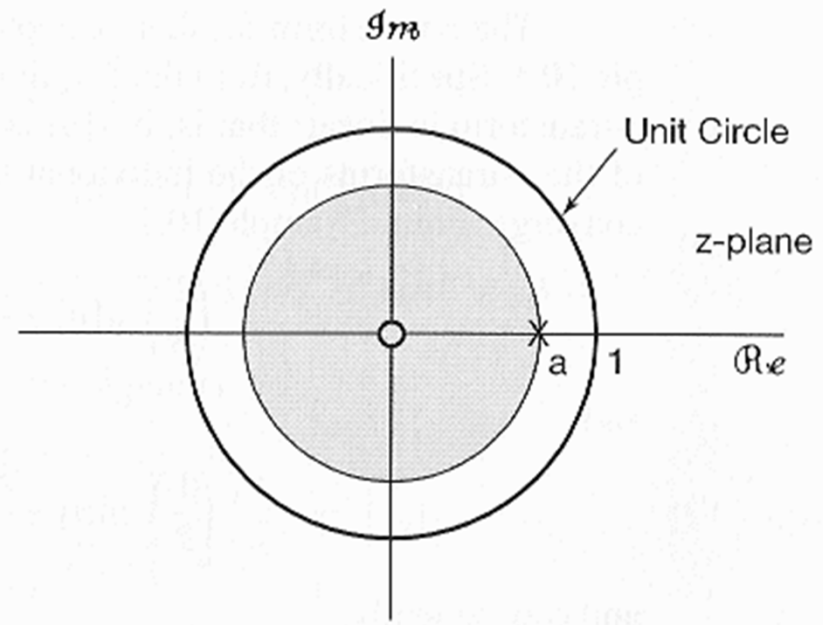
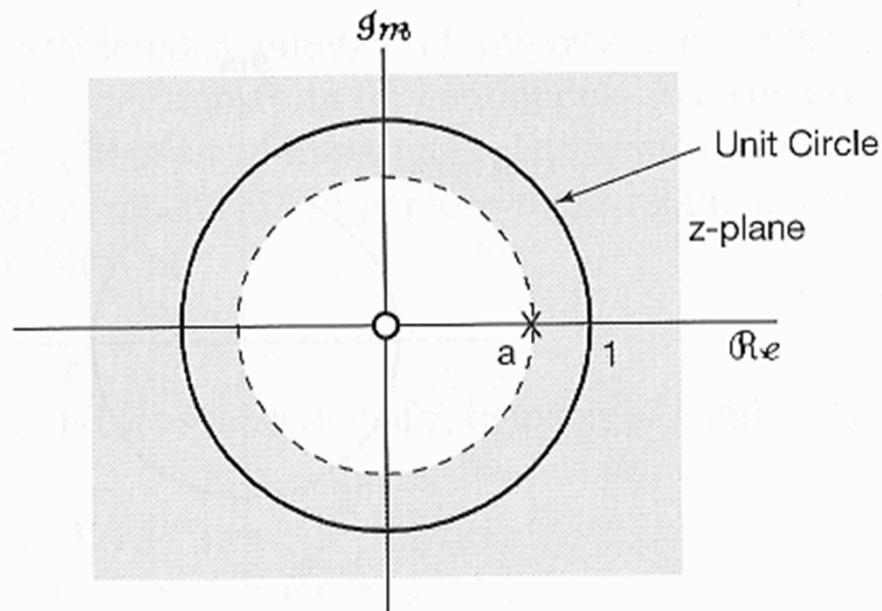


■ Region of Convergence (ROC):

$$a^n u[n] \xleftrightarrow{z} \frac{z}{z-a}, \quad |z| > |a|$$

$$-a^n u[-n-1] \xleftrightarrow{z} \frac{z}{z-a}, \quad |z| < |a|$$

where Fourier transform of $x[n]r^{-n}$ converges



■ Example 10.3:

$$x[n] = 7 \left(\frac{1}{3}\right)^n u[n] - 6 \left(\frac{1}{2}\right)^n u[n]$$

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{+\infty} \left\{ 7 \left(\frac{1}{3}\right)^n u[n] - 6 \left(\frac{1}{2}\right)^n u[n] \right\} z^{-n} \\ &= 7 \sum_{n=-\infty}^{+\infty} \left(\frac{1}{3}\right)^n u[n] z^{-n} - 6 \sum_{n=-\infty}^{+\infty} \left(\frac{1}{2}\right)^n u[n] z^{-n} \\ &= \frac{7}{1 - \frac{1}{3}z^{-1}} - \frac{6}{1 - \frac{1}{2}z^{-1}} \end{aligned}$$

$$7 \cdot \left(\frac{1}{3}\right)^n u[n] \xleftrightarrow{\mathcal{Z}} 7 \cdot \frac{1}{1 - \frac{1}{3}z^{-1}} = 7 \cdot \frac{z}{z - \frac{1}{3}}, \quad |z| > \frac{1}{3}$$

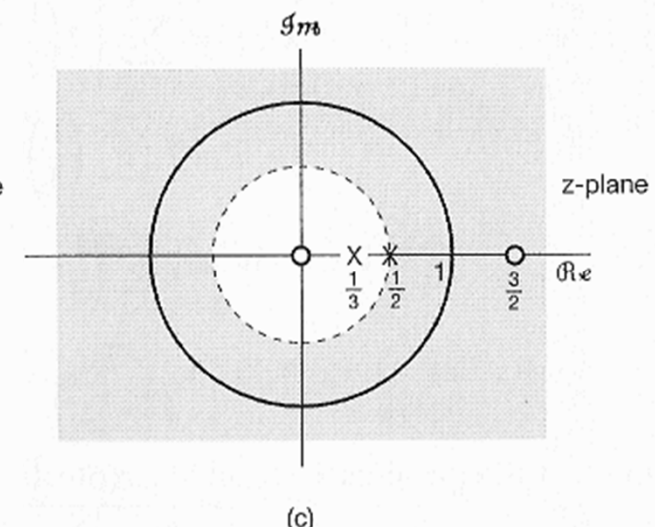
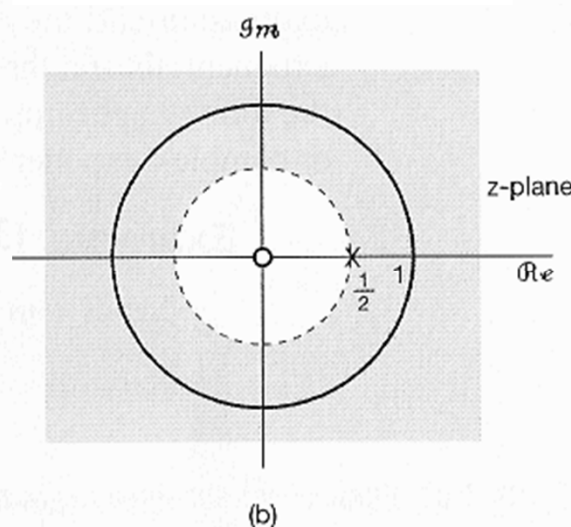
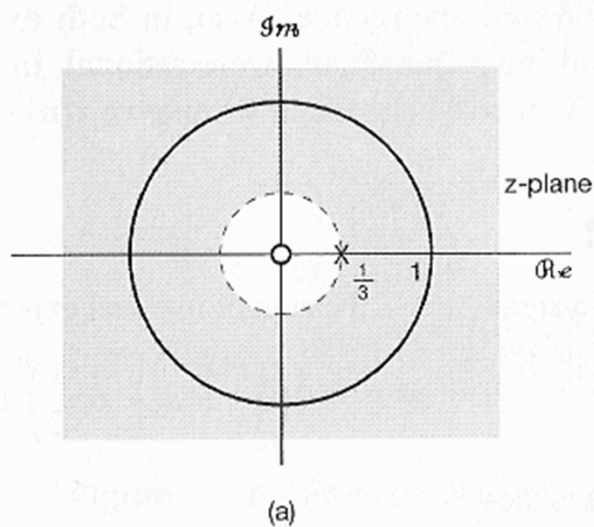
$$6 \cdot \left(\frac{1}{2}\right)^n u[n] \xleftrightarrow{\mathcal{Z}} 6 \cdot \frac{1}{1 - \frac{1}{2}z^{-1}} = 6 \cdot \frac{z}{z - \frac{1}{2}}, \quad |z| > \frac{1}{2}$$

Example 10.3:

$$7 \left(\frac{1}{3}\right)^n u[n] - 6 \left(\frac{1}{2}\right)^n u[n] \xleftrightarrow{z} \frac{7}{1 - \frac{1}{3}z^{-1}} - \frac{6}{1 - \frac{1}{2}z^{-1}}, \quad |z| > \frac{1}{2}$$

$$\xleftrightarrow{z} \frac{1 - \frac{3}{2}z^{-1}}{(1 - \frac{1}{3}z^{-1})(1 - \frac{1}{2}z^{-1})} \quad |z| > \frac{1}{2}$$

$$\xleftrightarrow{z} \frac{z(z - \frac{3}{2})}{(z - \frac{1}{3})(z - \frac{1}{2})}, \quad |z| > \frac{1}{2}$$



■ Example 10.4:

$$\left(\frac{1}{3}\right)^n \sin\left(\frac{\pi}{4}n\right) u[n]$$

$$\sin\left(\frac{\pi}{4}n\right) = \frac{1}{2j} \left(e^{j\pi/4n} - e^{-j\pi/4n} \right)$$

$$= \frac{1}{2j} \left(\left(e^{j\pi/4} \right)^n - \left(e^{-j\pi/4} \right)^n \right)$$

$$\left(\frac{1}{3}\right)^n \sin\left(\frac{\pi}{4}n\right) = \frac{1}{2j} \left(\left(\frac{1}{3} e^{j\pi/4} \right)^n - \left(\frac{1}{3} e^{-j\pi/4} \right)^n \right)$$

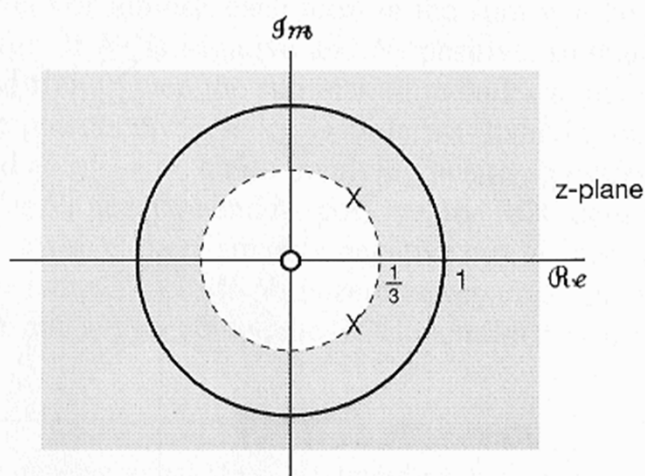
■ Example 10.4:

$$a^n u[n] \xleftrightarrow{z} \frac{z}{z - a}, \quad |z| > |a|$$

$$\left(\frac{1}{3} e^{j\pi/4}\right)^n u[n] \xleftrightarrow{z} \frac{z}{z - \frac{1}{3} e^{j\pi/4}}, \quad |z| > \frac{1}{3}$$

$$\left(\frac{1}{3} e^{-j\pi/4}\right)^n u[n] \xleftrightarrow{z} \frac{z}{z - \frac{1}{3} e^{-j\pi/4}}, \quad |z| > \frac{1}{3}$$

$$\left(\frac{1}{3}\right)^n \sin\left(\frac{\pi}{4}n\right) u[n] \xleftrightarrow{z} \frac{1}{2j} \left(\frac{z}{z - \frac{1}{3} e^{j\pi/4}} - \frac{z}{z - \frac{1}{3} e^{-j\pi/4}} \right), \quad |z| > \frac{1}{3}$$



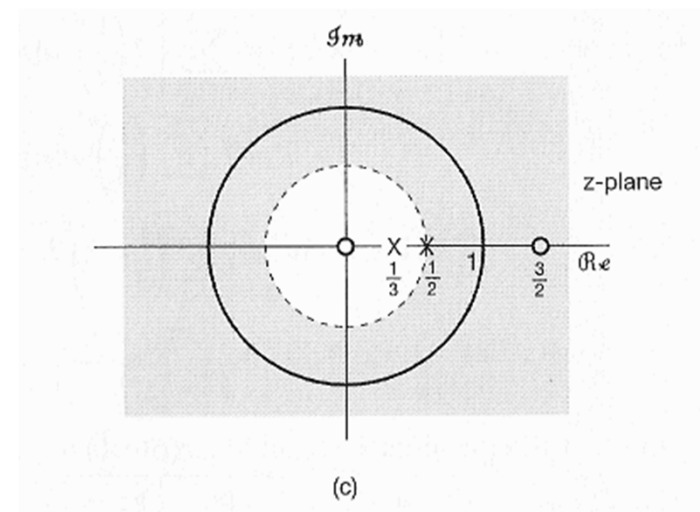
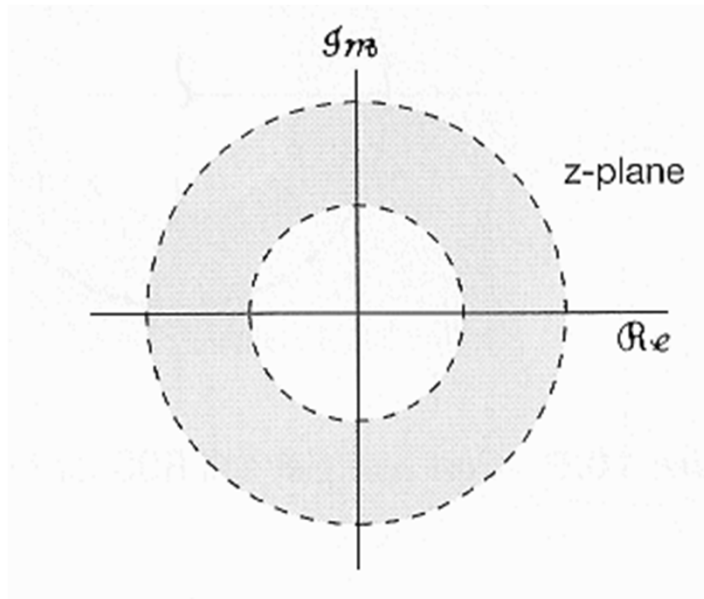
$$\frac{\frac{1}{3\sqrt{2}}z}{(z - \frac{1}{3} e^{j\pi/4})(z - \frac{1}{3} e^{-j\pi/4})}$$

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■ Properties of ROC:

1. The ROC of $X(z)$ consists of a ring in the z-plane centered about the origin
2. The ROC does not contain any poles

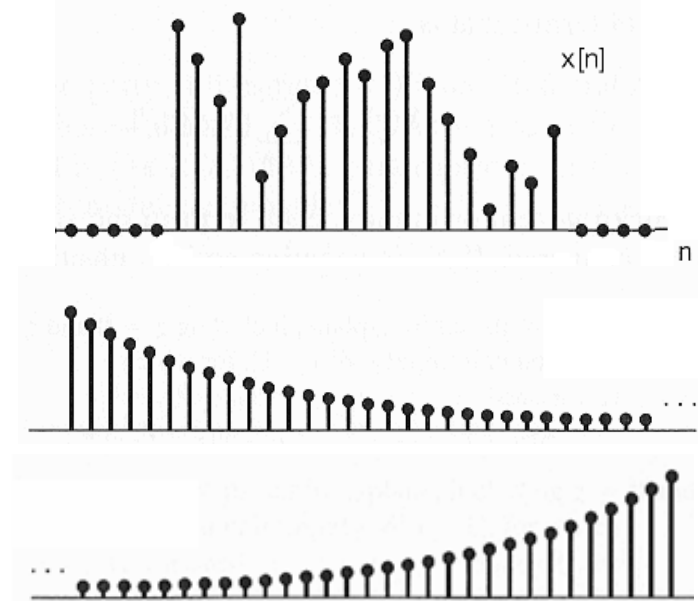
$$\frac{z(z - \frac{3}{2})}{(z - \frac{1}{3})(z - \frac{1}{2})}, \quad |z| > \frac{1}{2}$$



■ Properties of ROC:

3. If $x[n]$ is of finite duration,
then the ROC is the entire z-plane,
except possibly $z = 0$ and/or $z = \infty$

$$\begin{aligned}
 X(z) &\triangleq \sum_{n=-\infty}^{+\infty} x[n]z^{-n} \\
 &= \sum_{n=N_1}^{N_2} x[n]z^{-n} \text{ is bounded} \\
 &= \cdots + x[-3]z^{+3} + \cdots + x[5]z^{-5} \\
 &= \cdots + x[-3]z^{+3} + \cdots + x[5]\left(\frac{1}{z}\right)^5
 \end{aligned}$$



- However, $|z| \rightarrow 0 \Rightarrow |z|^N \rightarrow \infty$ if N is negative
- $|z| \rightarrow \infty \Rightarrow |z|^N \rightarrow \infty$ if N is positive

■ Properties of ROC:

4. If $x[n]$ is **right-sided** sequence, and if the circle $|z| = r_0$ is in the **ROC**, then all finite values of z for which $|z| > r_0$ will also be in the **ROC**

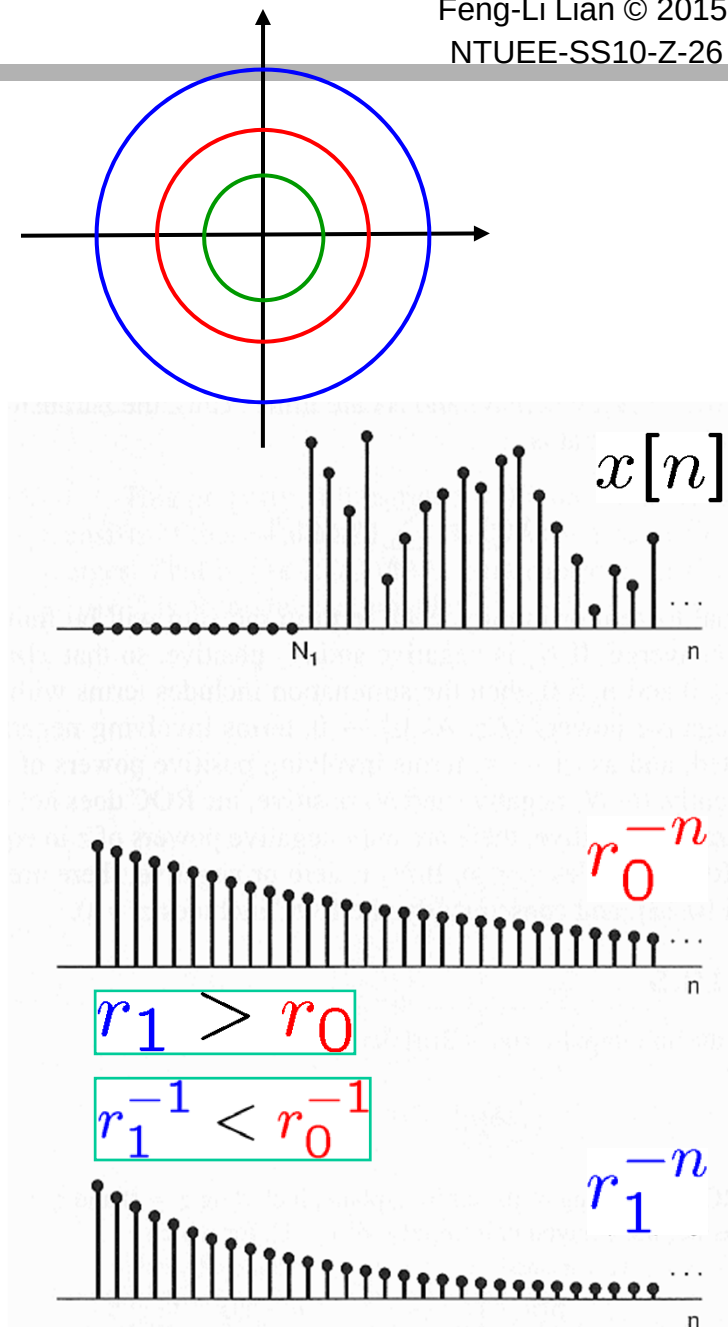
$$X(r_0 e^{j\omega}) = \sum_{n=N_1}^{\infty} \left\{ x[n] r_0^{-n} \right\} e^{-j\omega n} < \infty$$

$$X(r_1 e^{j\omega}) = \sum_{n=N_1}^{\infty} \left\{ x[n] r_1^{-n} \right\} e^{-j\omega n}$$

$$r_1 > r_0$$

$$r_1^{-1} < r_0^{-1}$$

$$< \sum_{n=N_1}^{\infty} \left\{ x[n] r_0^{-n} \right\} e^{-j\omega n} < \infty$$



■ Properties of ROC:

5. If $x[n]$ is left-sided sequence, and if the circle $|z| = r_0$ is in the ROC, then all values of z for which

$$0 < |z| < r_0$$

will also be in the ROC

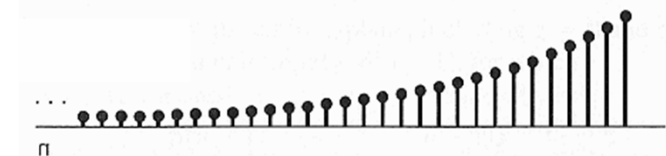
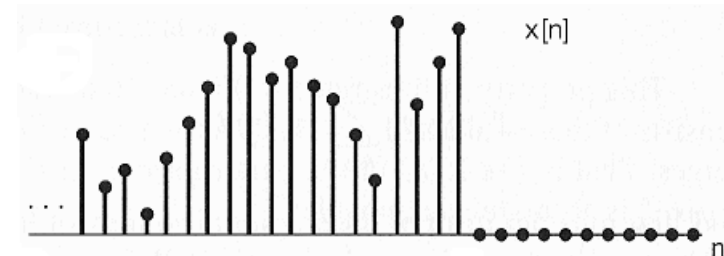
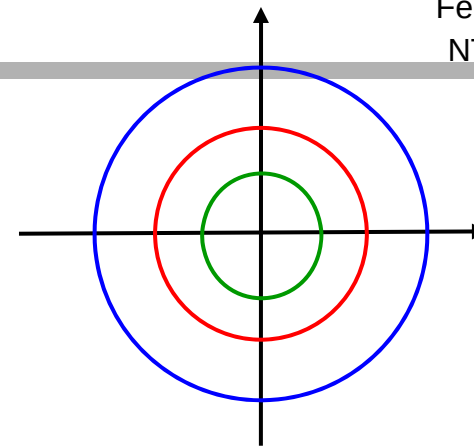
$$X(r_0 e^{j\omega}) = \sum_{n=-\infty}^{N_2} \left\{ x[n] r_0^{-n} \right\} e^{-j\omega n} < \infty$$

$$X(r_1 e^{j\omega}) = \sum_{n=-\infty}^{N_2} \left\{ x[n] r_1^{-n} \right\} e^{-j\omega n}$$

$$0 < r_1 < r_0$$

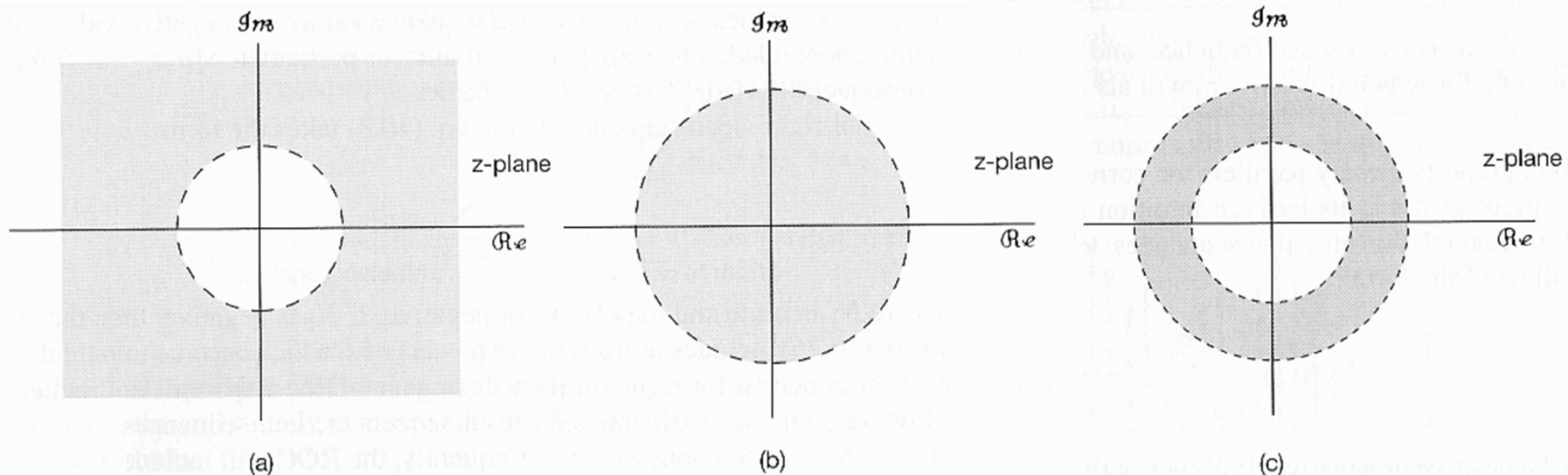
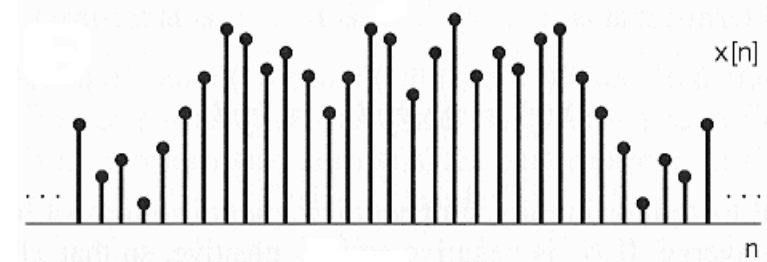
$$r_1^m < r_0^m, \quad m > 0$$

$$< \sum_{n=-\infty}^{N_2} \left\{ x[n] r_0^{-n} \right\} e^{-j\omega n} < \infty$$

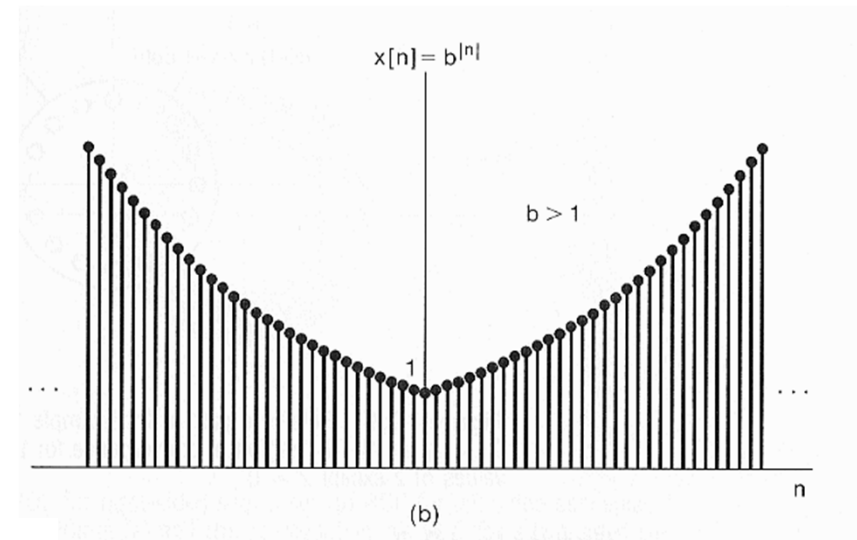
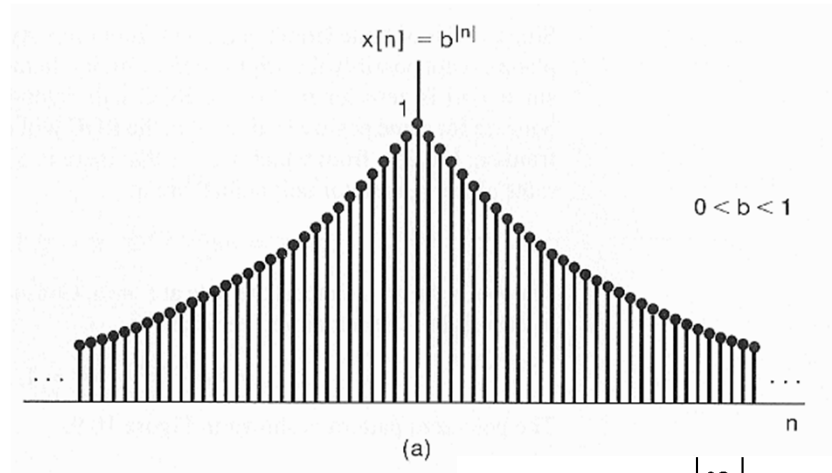


■ Properties of ROC:

6. If $x[n]$ is two-sided, and
if the circle $|z| = r_0$ is in the ROC,
then the ROC will consist of a ring
in the z-plane that includes the circle $|z| = r_0$



■ Example 10.7:



$$x[n] = b^{|n|}, \quad b > 0$$

$$= b^n u[n] + b^{-n} u[-n-1]$$

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n] z^{-n}$$

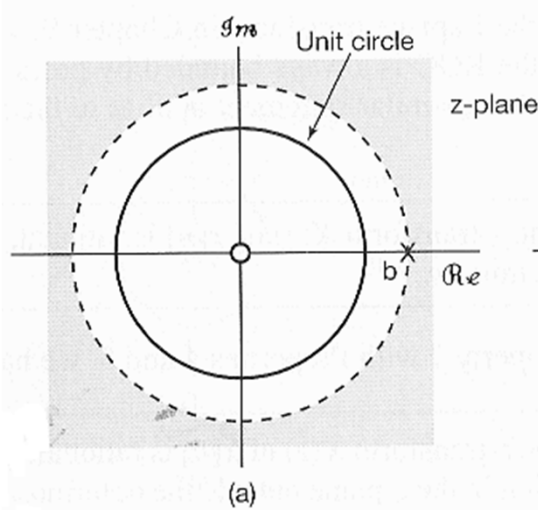
$$X(z) = \frac{1}{1 - bz^{-1}} - \frac{1}{1 - b^{-1}z^{-1}}, \quad b < |z| < \frac{1}{b}$$

$$= \left(\frac{b^2 - 1}{b} \right) \frac{z}{(z - b)(z - b^{-1})}, \quad b < |z| < \frac{1}{b}$$

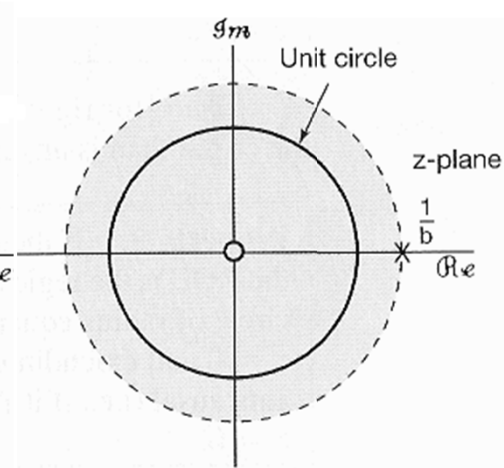
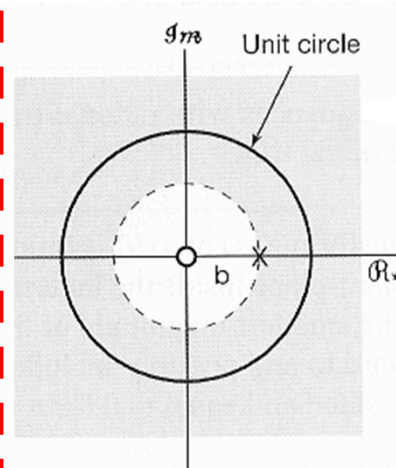
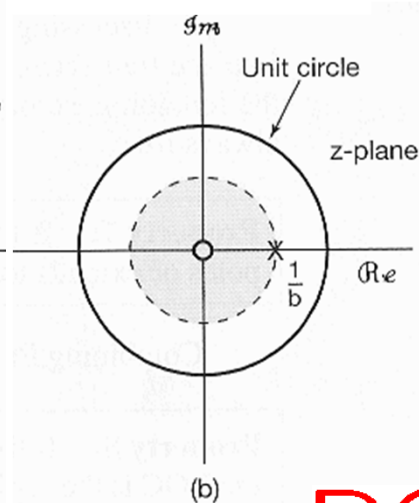
■ Example 10.7:

$$X(z) = \left(\frac{b^2 - 1}{b} \right) \frac{z}{(z - b)(z - b^{-1})}, \quad b < |z| < \frac{1}{b}$$

$b > 1$

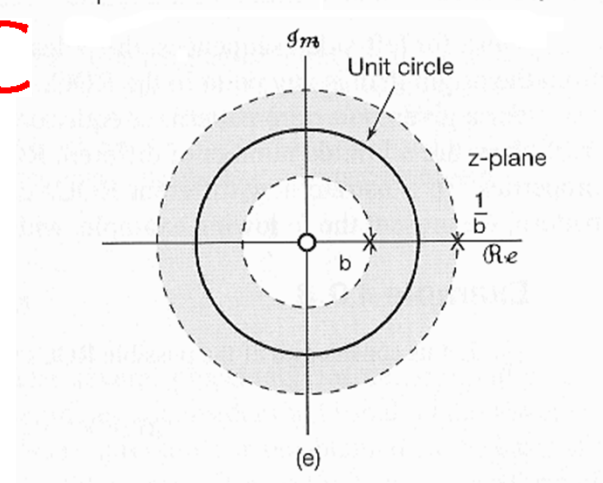


$0 < b < 1$



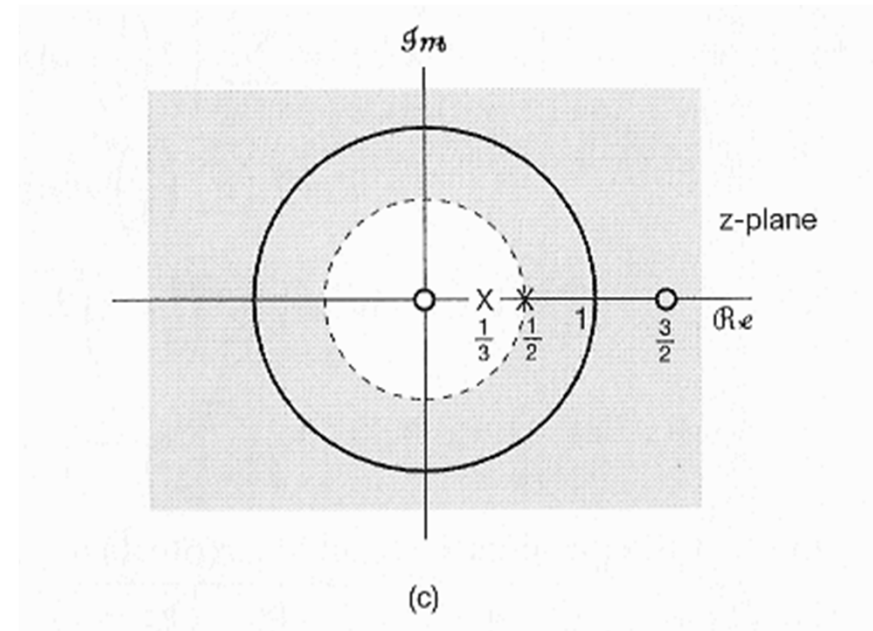
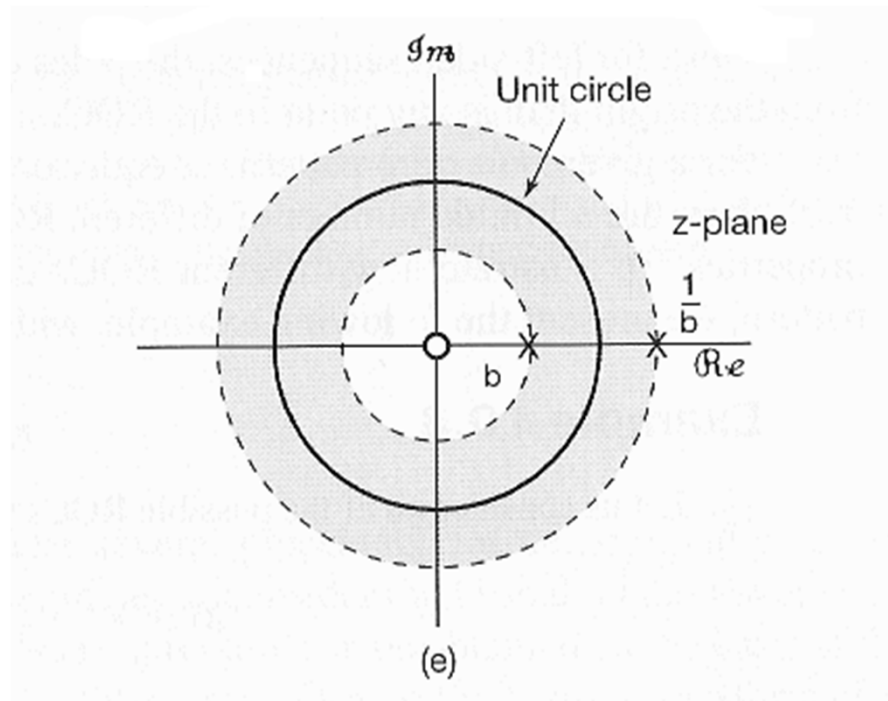
ROC

X



■ Properties of ROC:

7. If the z-transform $X(z)$ of $x[n]$ is rational,
then its **ROC** is bounded by poles or extends to ∞



$$X(z) = \left(\frac{b^2 - 1}{b} \right) \frac{z}{(z - b)(z - b^{-1})}, \quad b < |z| < \frac{1}{b}$$

$$\frac{z(z - \frac{3}{2})}{(z - \frac{1}{3})(z - \frac{1}{2})}, \quad |z| > \frac{1}{2}$$

■ Properties of ROC:

8. If the z-transform $X(z)$ of $x[n]$ is rational

– If $x[n]$ is right sided,

then the ROC is the region in the z-plane

outside the outermost pole ---

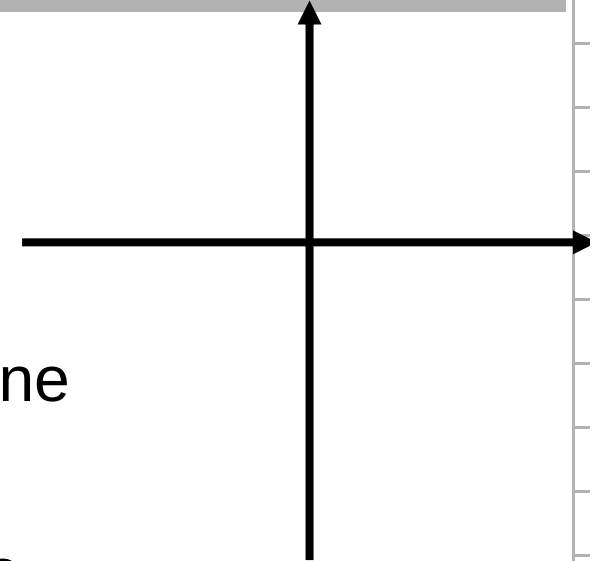
i.e., outside the circle of radius equal to

the largest magnitude of the poles of $X(z)$

– Furthermore, if $x[n]$ is causal,

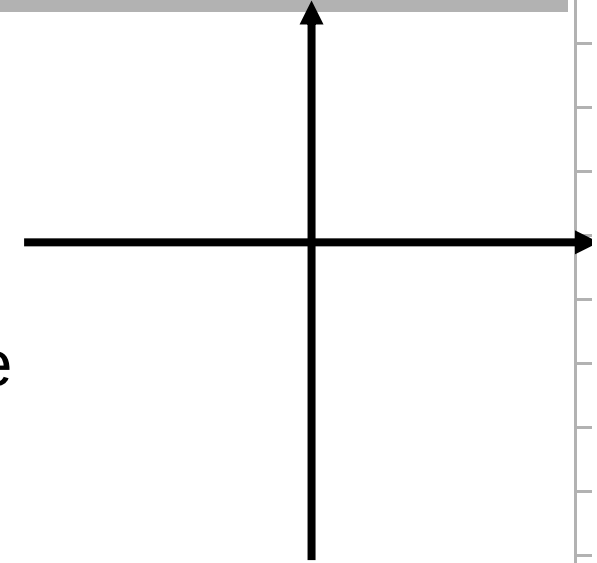
then the ROC also includes $z = \infty$

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n] z^{-n} = \sum_{n=0}^{+\infty} x[n] \left(\frac{1}{z}\right)^n$$



■ Properties of ROC:

9. If the z-transform $X(z)$ of $x[n]$ is rational and if $x[n]$ is left sided, then the ROC is the region in the z-plane inside the innermost pole --- i.e., inside the circle of radius equal to the smallest magnitude of the poles of $X(z)$ other than any at $z = 0$ and extending inward and possibly including $z = 0$
- In particular, if $x[n]$ is anti-causal, (i.e., if it is left sided and $= 0$ for $n > 0$), then the ROC also includes $z = 0$

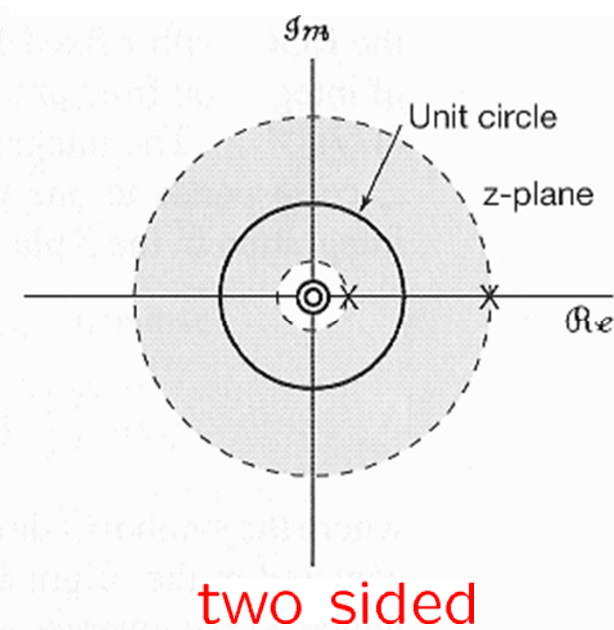
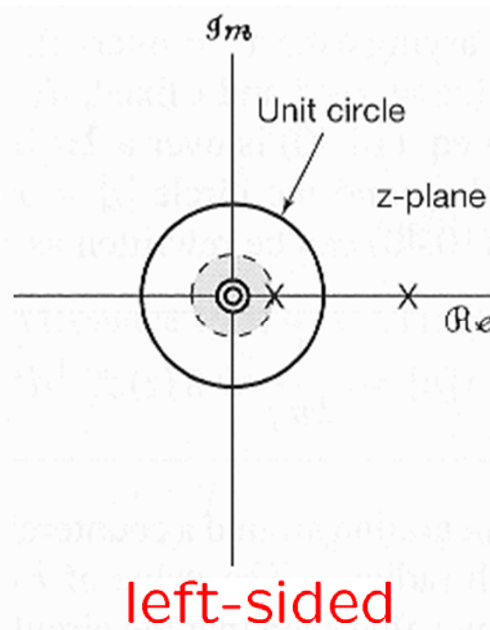
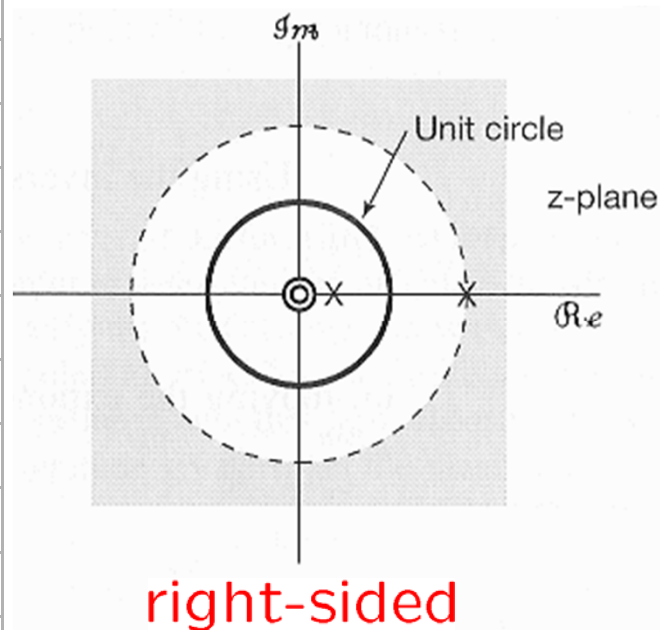
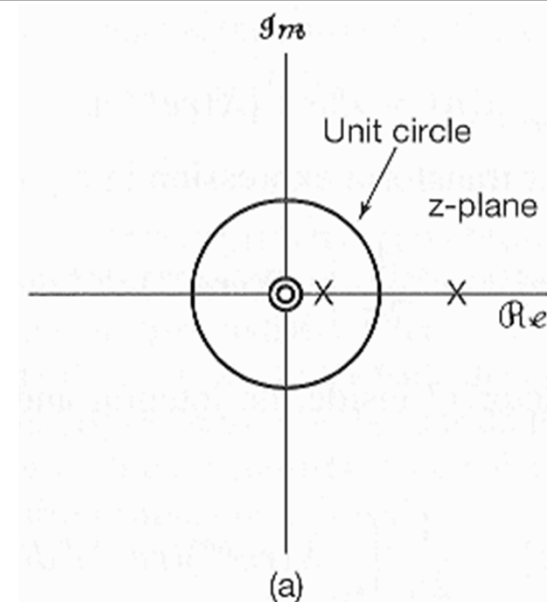


$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{+\infty} x[n] z^{-n} \\ &= \sum_{n=-\infty}^0 x[n] z^{-n} \\ &= \sum_{m=0}^{\infty} x[-m] z^m \end{aligned}$$

■ Example 10.8:

$$X(z) = \frac{1}{(1 - \frac{1}{3}z^{-1})(1 - 2z^{-1})}$$

$$= \frac{z^2}{(z - \frac{1}{3})(z - 2)}$$



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■ The Inverse z-Transform:

- By the use of contour integration

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n]z^{-n}$$

$$X(re^{jw}) = \mathcal{F} \left\{ x[n]r^{-n} \right\}$$

$\forall z = re^{jw}$ in the ROC

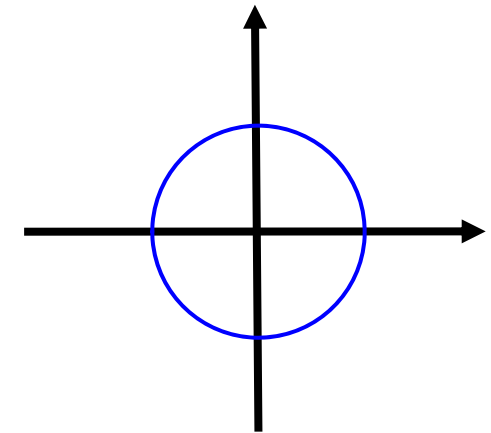
$$x[n]r^{-n} = \mathcal{F}^{-1} \left\{ X(re^{jw}) \right\}$$

$$x[n] = r^n \mathcal{F}^{-1} \left\{ X(re^{jw}) \right\}$$

$$= r^n \frac{1}{2\pi} \int_{2\pi} X(re^{jw}) e^{jwn} dw$$

$$= \frac{1}{2\pi} \int_{2\pi} X(re^{jw}) (re^{jw})^n dw$$

$$\Rightarrow x[n] = \frac{1}{2\pi j} \oint X(z) z^{n-1} dz$$



$$z = re^{jw}$$

$$dz = jre^{jw} dw = jz dw$$

$$dw = \frac{1}{jz} dz$$

■ The Inverse z-Transform:

$$a^n u[n] \xleftrightarrow{z} \frac{z}{z - a} = \frac{1}{1 - az^{-1}} \quad |z| > |a|$$

$$-a^n u[-n - 1] \xleftrightarrow{z} \frac{z}{z - a} = \frac{1}{1 - az^{-1}} \quad |z| < |a|$$

- By the technique of partial fraction expansion

$$X(z) = \frac{A}{1 - az^{-1}} + \frac{B}{1 - bz^{-1}} + \cdots + \frac{M}{1 - mz^{-1}}$$

$$x[n] = A a^n u[n] - B b^n u[-n - 1] + \cdots + x_m[n]$$

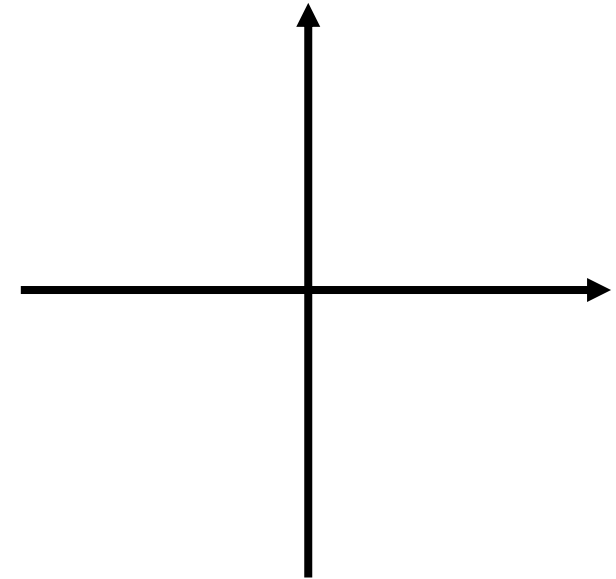
(if ROC outside $z = a$) (if ROC inside $z = b$)

■ Example 10.9:

$$X(z) = \frac{3 - \frac{5}{6}z^{-1}}{(1 - \frac{1}{4}z^{-1})(1 - \frac{1}{3}z^{-1})}, \quad |z| > \frac{1}{3}$$

$$= \frac{3z(z - \frac{5}{18})}{(z - \frac{1}{4})(z - \frac{1}{3})}$$

$$X(z) = \frac{1}{(1 - \frac{1}{4}z^{-1})} + \frac{2}{(1 - \frac{1}{3}z^{-1})}$$



$$\left(\frac{1}{4}\right)^n u[n] \xleftrightarrow{z} \frac{1}{(1 - \frac{1}{4}z^{-1})}, \quad |z| > \frac{1}{4}$$

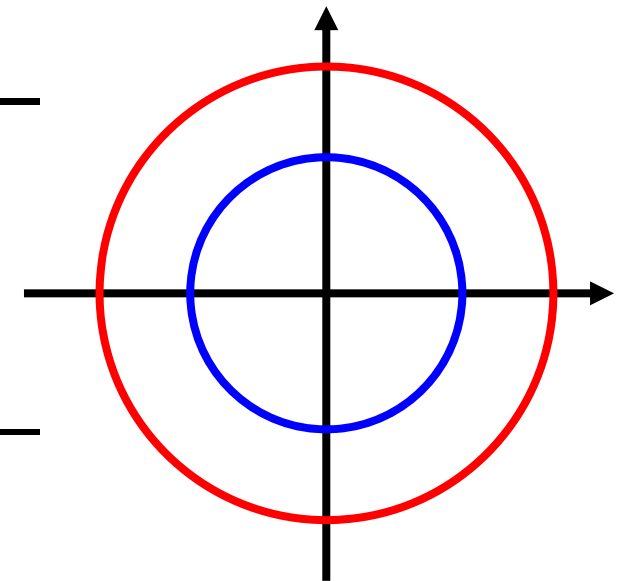
$$\left(\frac{1}{3}\right)^n u[n] \xleftrightarrow{z} \frac{1}{(1 - \frac{1}{3}z^{-1})}, \quad |z| > \frac{1}{3}$$

$$\Rightarrow x[n] = \left(\frac{1}{4}\right)^n u[n] + 2 \left(\frac{1}{3}\right)^n u[n]$$

■ Examples 10.9, 10.10, 10.11:

	$ z < \frac{1}{4}$	$\frac{1}{4} < z $
$\frac{1}{(1 - \frac{1}{4}z^{-1})}$	$-\left(\frac{1}{4}\right)^n u[-n-1]$	$\left(\frac{1}{4}\right)^n u[n]$

	$ z < \frac{1}{3}$	$\frac{1}{3} < z $
$\frac{1}{(1 - \frac{1}{3}z^{-1})}$	$-\left(\frac{1}{3}\right)^n u[-n-1]$	$\left(\frac{1}{3}\right)^n u[n]$



	$ z < \frac{1}{4}$	$\frac{1}{4} < z < \frac{1}{3}$	$\frac{1}{3} < z $
$\frac{1}{(1 - \frac{1}{4}z^{-1})}$	$-\left(\frac{1}{4}\right)^n u[-n-1]$	$\left(\frac{1}{4}\right)^n u[n]$	$\left(\frac{1}{4}\right)^n u[n]$
$\frac{1}{(1 - \frac{1}{3}z^{-1})}$	$-\left(\frac{1}{3}\right)^n u[-n-1]$	$-\left(\frac{1}{3}\right)^n u[-n-1]$	$\left(\frac{1}{3}\right)^n u[n]$

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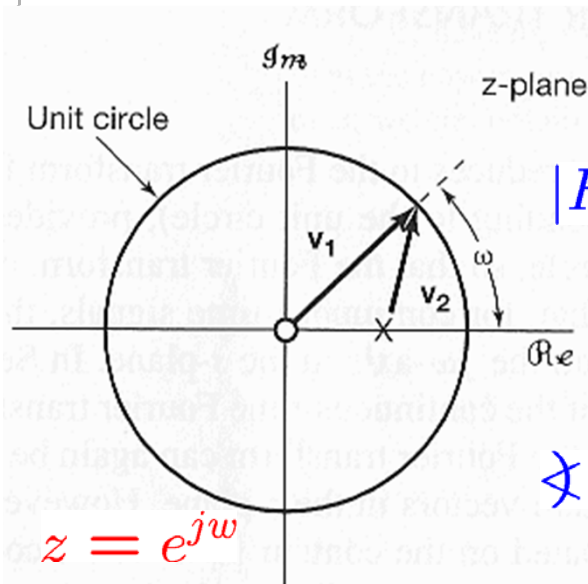
First-Order Systems:

$$h[n] = a^n u[n]$$

$$H(z) = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}, \quad |z| > |a|$$

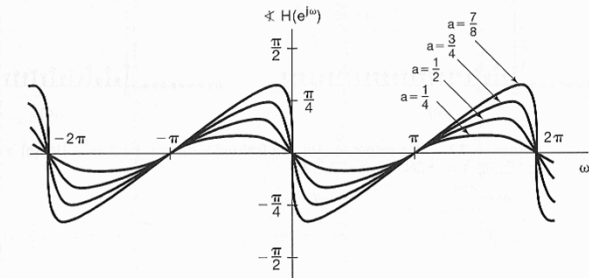
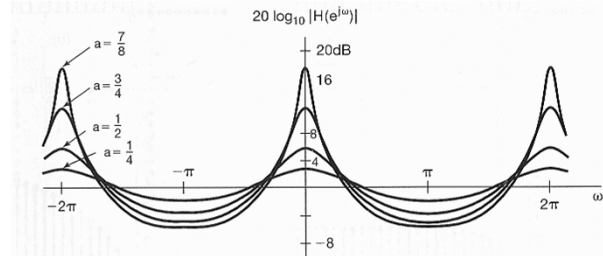
- For $|a| < 1$, ROC includes $|z| = 1$

$$\Rightarrow H(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}}$$

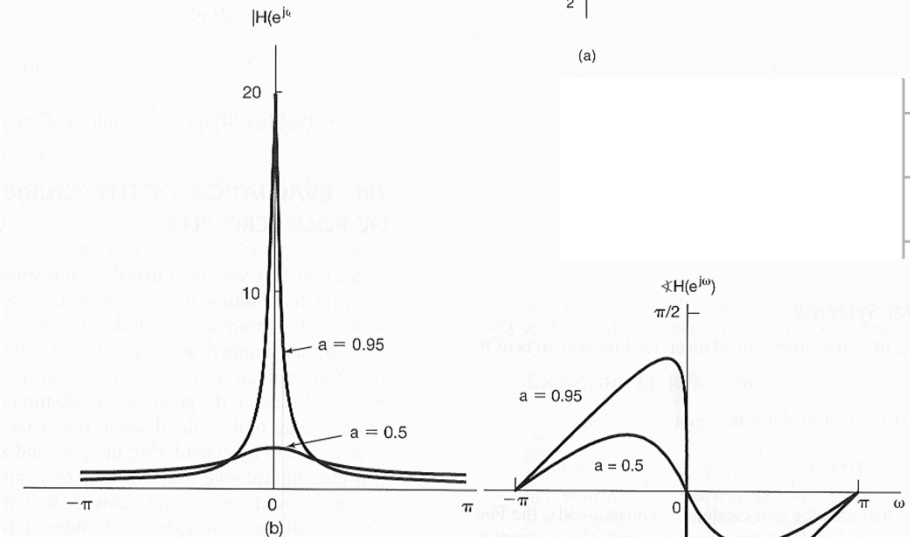


$$|H| = \frac{1}{|1 - ae^{-j\omega}|}$$

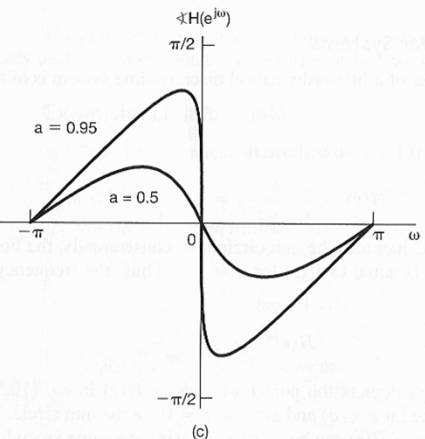
$$\angle H = \angle \frac{1}{1 - ae^{-j\omega}}$$



(a)



(b)



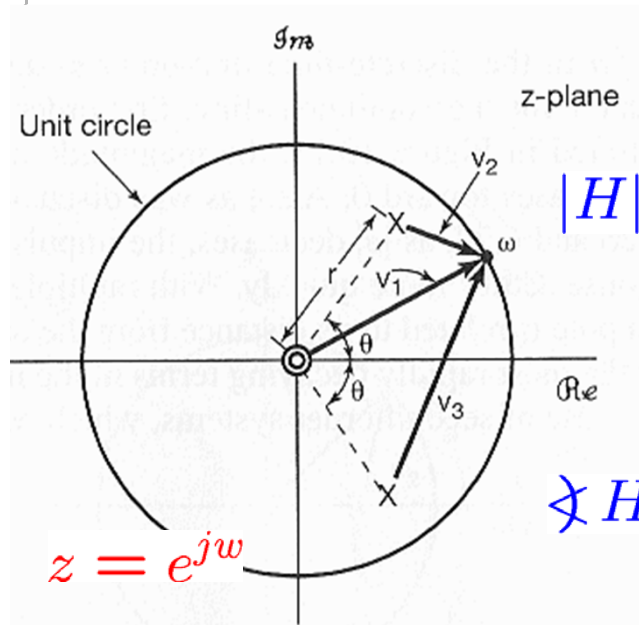
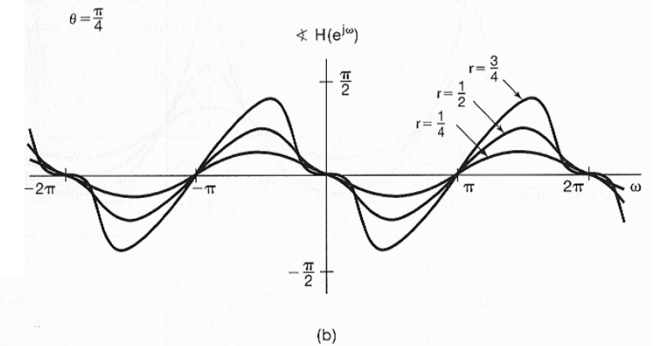
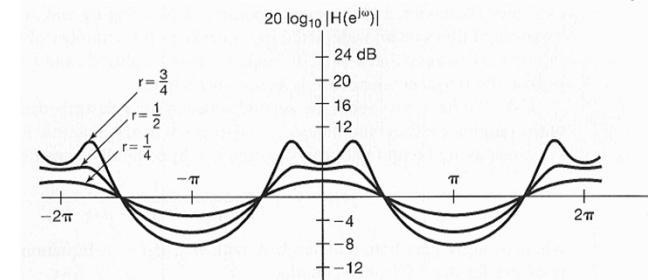
(c)

Second-Order Systems:

$$H(z) = \frac{1}{1 - (2r \cos \theta)z^{-1} + (r^2)z^{-2}}$$

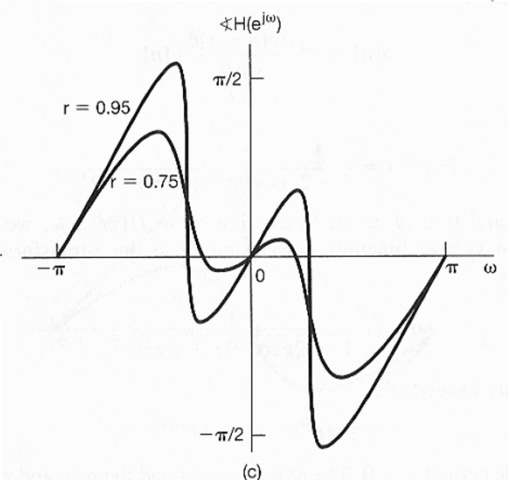
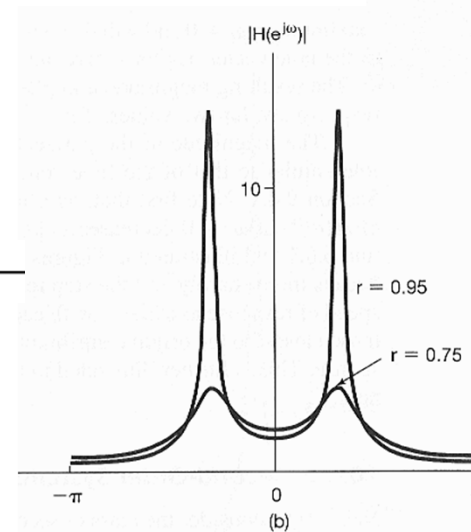
$$= \frac{z^2}{(z - p_1)(z - p_2)}, \quad |z| > |r|$$

poles : $\begin{cases} p_1 = re^{j\theta} \\ p_2 = re^{-j\theta} \end{cases}$ zeros : $z_1 = z_2 = 0$



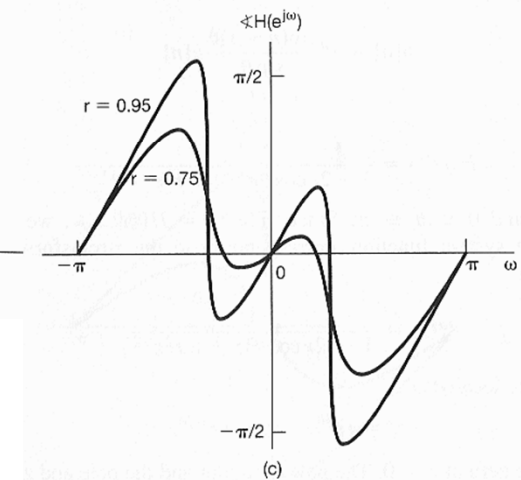
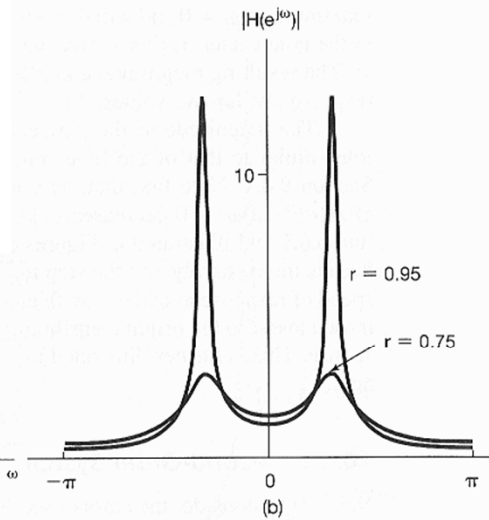
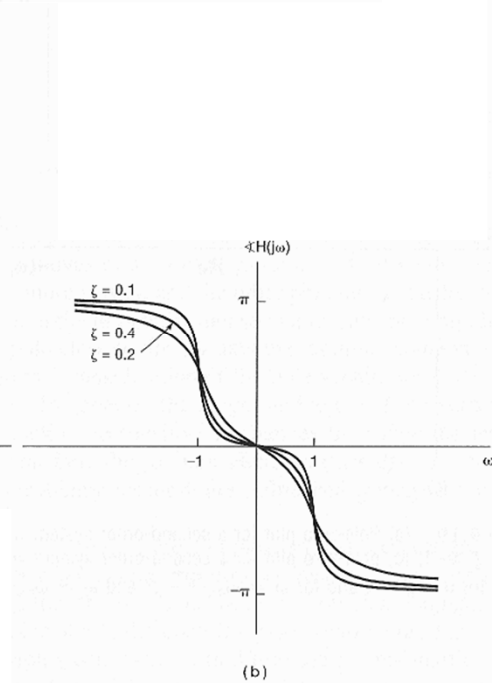
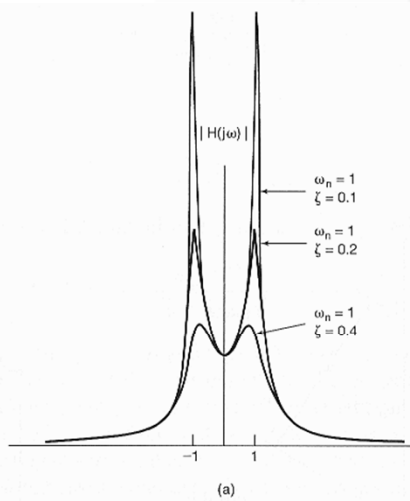
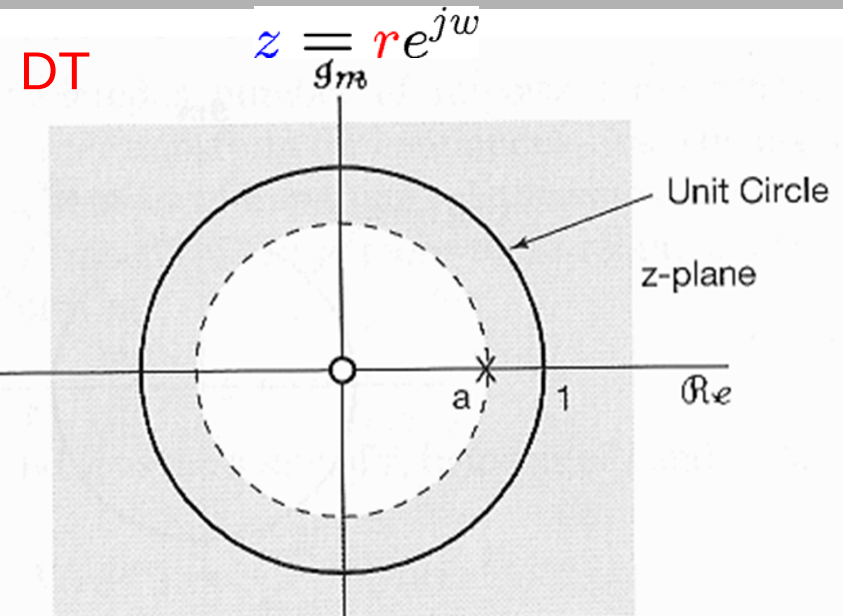
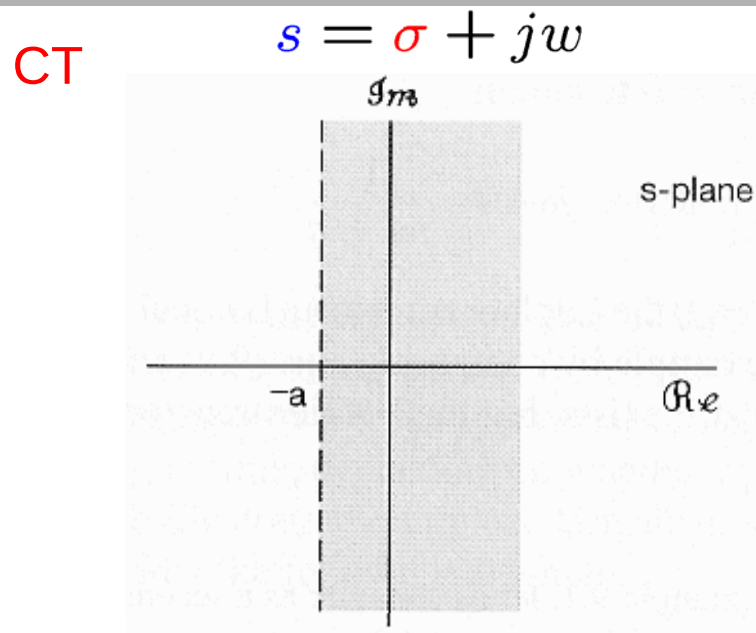
$$|H| =$$

$$\angle H =$$



Laplace Transform and The z-Transform

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NTUEE-SS10-Z-48



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Property	CTFS	DTFS	CTFT	DTFT	LT	zT
Linearity	3.5.1		4.3.1	5.3.2	9.5.1	10.5.1
Time Shifting	3.5.2		4.3.2	5.3.3	9.5.2	10.5.2
Frequency Shifting (in s, z)			4.3.6	5.3.3	9.5.3	10.5.3
Conjugation	3.5.6		4.3.3	5.3.4	9.5.5	10.5.6
Time Reversal	3.5.3		4.3.5	5.3.6		10.5.4
Time & Frequency Scaling	3.5.4		4.3.5	5.3.7	9.5.4	10.5.5
(Periodic) Convolution			4.4	5.4	9.5.6	10.5.7
Multiplication	3.5.5	3.7.2	4.5	5.5		
Differentiation/First Difference		3.7.2	4.3.4, 4.3.6	5.3.5, 5.3.8	9.5.7, 9.5.8	10.5.7, 10.5.8
Integration/Running Sum (Accumulation)			4.3.4	5.3.5	9.5.9	10.5.7
Conjugate Symmetry for Real Signals	3.5.6		4.3.3	5.3.4		
Symmetry for Real and Even Signals	3.5.6		4.3.3	5.3.4		
Symmetry for Real and Odd Signals	3.5.6		4.3.3	5.3.4		
Even-Odd Decomposition for Real Signals			4.3.3	5.3.4		
Parseval's Relation for (A)Periodic Signals	3.5.7	3.7.3	4.3.7	5.3.9		
Initial- and Final-Value Theorems					9.5.10	10.5.9

■ Linearity of the z-Transform:

$$x_1[n] \xleftrightarrow{z} X_1(z), \quad ROC = R_1$$

$$x_2[n] \xleftrightarrow{z} X_2(z), \quad ROC = R_2$$

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n] z^{-n}$$

$$x[n] = \frac{1}{2\pi j} \oint X(z) z^{n-1} dz$$

$$ax_1[n] + bx_2[n] \xleftrightarrow{z} aX_1(z) + bX_2(z),$$

with ROC containing $\underline{R_1 \cap R_2}$

$$\sum_{n=-\infty}^{+\infty} a x_1[n] z^{-n} + \sum_{n=-\infty}^{+\infty} b x_2[n] z^{-n}$$

$$a \left(\sum_{n=-\infty}^{+\infty} x_1[n] z^{-n} \right) + b \left(\sum_{n=-\infty}^{+\infty} x_2[n] z^{-n} \right)$$

$$a X_1 \left(\quad \right) + b X_2 \left(\quad \right)$$

■ Time Shifting:

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n] z^{-n}$$

$$x[n] = \frac{1}{2\pi j} \oint X(z) z^{n-1} dz$$

$$x[n] \xleftrightarrow{z} X(z), \quad ROC = R$$

$$\sum_{n=-\infty}^{+\infty} x[n-n_0] z^{-n} \xleftrightarrow{z} z^{-n_0} X(z), \quad ROC = R$$

$m = n - n_0 \quad n = m + n_0$

$$\sum_{m=-\infty}^{+\infty} x[m] z^{-(m+n_0)}$$

except for the possible
addition or deletion of
the origin or infinity

$$= \left(\sum_{m=-\infty}^{+\infty} x[m] z^{-(m)} \right) z^{-n_0} = z^{-n_0} X(z)$$

$$= z^{-n_0} X(z)$$

■ Scaling in the z-Domain:

$$x[n] \xleftrightarrow{z} X(z), \quad \text{ROC} = R$$

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n] z^{-n}$$

$$x[n] = \frac{1}{2\pi j} \oint X(z) z^{n-1} dz$$

$$\sum_{n=-\infty}^{+\infty} z_0^n x[n] z^{-n} \xleftrightarrow{z} X\left(\frac{z}{z_0}\right), \quad \text{ROC} = |z_0| R$$

$$\sum_{n=-\infty}^{+\infty} z_0^n x[n] z^{-n}$$

$$\frac{1}{2\pi j} \oint X\left(\frac{z}{z_0}\right) z^{n-1} dz$$

$$\sum_{n=-\infty}^{+\infty} x[n] \left(\frac{z}{z_0}\right)^{-n}$$

$$\frac{1}{2\pi j} \oint X\left(\frac{z}{z_0}\right) \left(\frac{z}{z_0}\right)^{n-1} dz$$

$$\sum_{n=-\infty}^{+\infty} x[n] \left(\frac{z}{z_0}\right)^{-n}$$

$$\frac{1}{2\pi j} \oint X\left(\frac{z}{z_0}\right) \left(\frac{z}{z_0}\right)^{n-1} dz$$

$$\frac{1}{2\pi j} \oint X\left(\frac{z}{z_0}\right) \left(\frac{z}{z_0}\right)^{n-1} dz$$

$$X\left(\frac{z}{z_0}\right) = X\left(\frac{z}{z_0}\right)$$

$$x[n]$$

Scaling in the z-Domain:

$$a^n u[n] \xleftrightarrow{z} X(z) = \frac{z}{z - a}, \quad |z| > |a|$$

$(\frac{1}{2})^n$

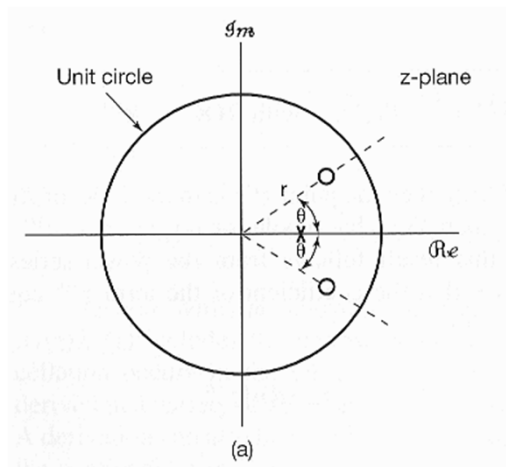
$$(b a)^n u[n] \xleftrightarrow{z} X\left(\frac{z}{b}\right) = \frac{\frac{z}{b}}{\frac{z}{b} - a}, \quad \left|\frac{z}{b}\right| > |a|$$

$(4 \frac{1}{2})^n$
 $= (2)^n$

$|z| > |b| |a|$
 $|z| > 4 \cdot \frac{1}{2} = 2$

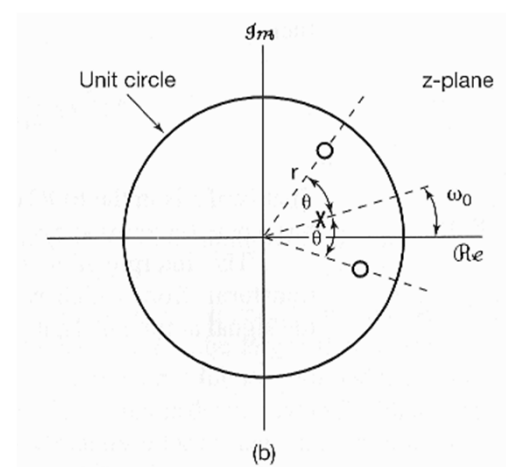
$$e^{j\omega_0 n} x[n] \xleftrightarrow{z} X(e^{-j\omega_0} z), \quad ROC = R$$

$(e^{j\omega_0})^n$



$a e^{jb}$

$a e^{jb}$



$a e^{jb} e^{j\omega_0}$

$a e^{jb} e^{j\omega_0}$

Time Reversal:

$$x[n] \xleftrightarrow{z} X(z), \quad \text{ROC} = R$$

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n] z^{-n}$$

$$x[n] = \frac{1}{2\pi j} \oint X(z) z^{n-1} dz$$

$$\sum_{n=-\infty}^{+\infty} x[-n] z^{-n} \xleftrightarrow{z} X\left(\frac{1}{z}\right), \quad \text{ROC} = \frac{1}{R}$$

$=$

$$\sum_{m=-\infty}^{+\infty} x[m] (z)^{+m}$$

$$\sum_{m=-\infty}^{+\infty} x[m] \left(\frac{1}{z}\right)^{-m}$$

$$X\left(\frac{1}{z}\right)$$

Time Expansion:

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n]z^{-n}$$

$$x[n] = \frac{1}{2\pi j} \oint X(z)z^{n-1}dz$$

$$x_{(k)}[n] = \begin{cases} x[n/k], & \text{if } n = k \cdot m \\ 0, & \text{otherwise} \end{cases} \quad \begin{matrix} k \text{ is a constant} \\ m \text{ is a new time variable} \end{matrix}$$

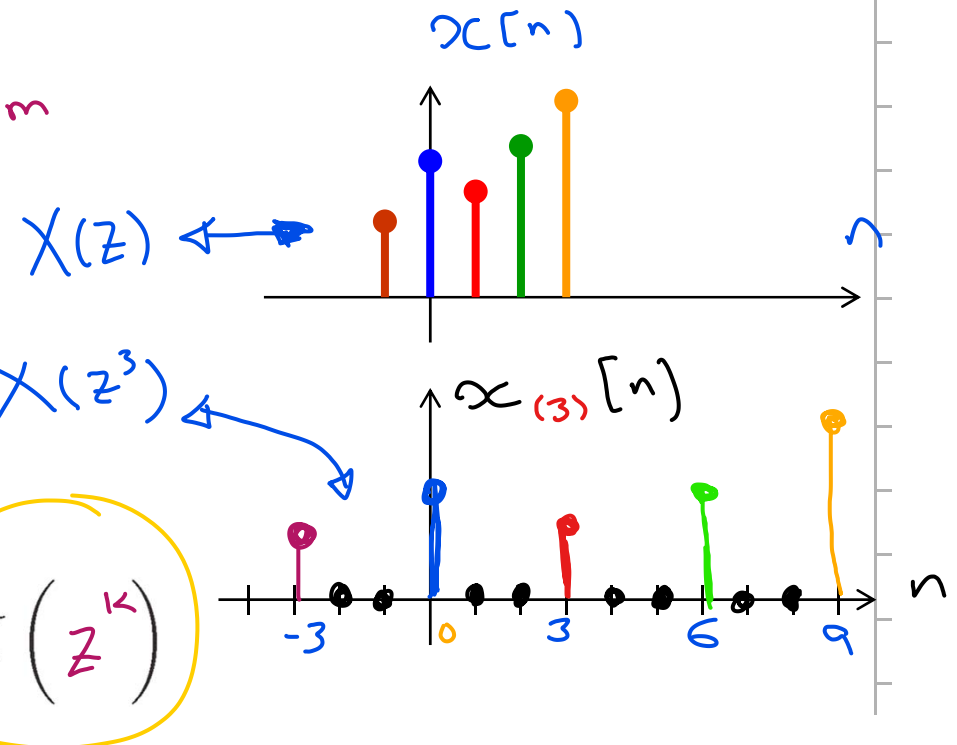
$$x_{(k)}[n] \xleftrightarrow{z} X(z^k), \quad ROC = R^{1/k}$$

$$\sum_{n=-\infty}^{+\infty} x\left[\frac{n}{k}\right] z^{-n}$$

$$n = k \cdot m$$

$$\sum_{m=-\infty}^{+\infty} x\left[\frac{k \cdot m}{k}\right] z^{-(k \cdot m)}$$

$$\sum_{m=-\infty}^{+\infty} x[m] (z^k)^{-m}$$



■ Conjugation:

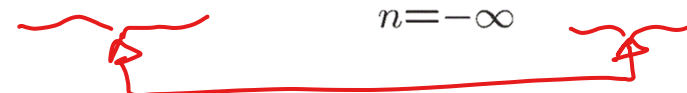
$$x[n] \xleftrightarrow{z} X(z), \quad ROC = R$$

$$x^*[n] \xleftrightarrow{z} X^*(z^*), \quad ROC = R$$

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n] z^{-n}$$

$$x[n] = \frac{1}{2\pi j} \oint X(z) z^{n-1} dz$$

$$\begin{aligned} (X(z))^* &= \left(\sum_{n=-\infty}^{+\infty} x[n] z^{-n} \right)^* \\ &= \sum_{n=-\infty}^{+\infty} x^*[n] (z^{-n})^* \\ &= \sum_{n=-\infty}^{+\infty} x^*[n] (z^*)^{-n} \end{aligned}$$

$$X^*(z^*) = \sum_{n=-\infty}^{+\infty} x^*[n] (z^*)^{-n}$$


■ Convolution Property:

$$x_1[n] \xleftrightarrow{z} X_1(z), \quad \text{ROC} = R_1$$

$$x_2[n] \xleftrightarrow{z} X_2(z), \quad \text{ROC} = R_2$$

$$x_1[n] * x_2[n] \xleftrightarrow{z} X_1(z)X_2(z),$$

$$= \sum_{n=-\infty}^{+\infty} \left(\sum_{m=-\infty}^{+\infty} x_1[m] x_2[n-m] \right) z^{-n}$$

$$= \sum_{m=-\infty}^{+\infty} x_1[m] \left(\sum_{n=-\infty}^{+\infty} x_2[n-m] z^{-n} \right)$$

$$= \sum_{m=-\infty}^{+\infty} x_1[m] \left(\sum_{k=-\infty}^{+\infty} x_2[k] z^{-(k+m)} \right)$$

$$= \left(\sum_{m=-\infty}^{+\infty} x_1[m] z^{-m} \right) \left(\sum_{k=-\infty}^{+\infty} x_2[k] z^{-k} \right) \quad X_1(z) X_2(z)$$

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n] z^{-n}$$

$$x[n] = \frac{1}{2\pi j} \oint X(z) z^{n-1} dz$$

with ROC containing $R_1 \cap R_2$

$R_1 \cap R_2$ may be larger
if pole-zero cancellation
occurs in the product

■ Differentiation in the z-Domain:

$$x[n] \xleftrightarrow{z} X(z), \quad ROC = R$$

$$nx[n] \xleftrightarrow{z} -z \frac{d}{dz} X(z), \quad ROC = R$$

$$\frac{d}{dz} X(z) = \frac{d}{dz} \left(\sum_{n=-\infty}^{+\infty} x[n] z^{-n} \right)$$

$$= \sum_{n=-\infty}^{+\infty} x[n] \frac{d}{dz} (z^{-n})$$

$$\frac{d}{dz} X(z) = \sum_{n=-\infty}^{+\infty} x[n] (-n) z^{-n-1}$$

$$(-z) \frac{d}{dz} X(z) = \sum_{n=-\infty}^{+\infty} x[n] n z^{-n}$$

$$(-z) \frac{d}{dz} X(z) = \sum_{n=-\infty}^{+\infty} (n x[n]) z^{-n}$$

If $x[n] = 0$ for $n < 0$

■ The Initial-Value Theorem:

$$\Rightarrow x[0] = \lim_{z \rightarrow \infty} X(z)$$

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n] z^{-n} = x[0] + x[1] z^{-1} + x[2] z^{-2}$$

■ The Final-Value Theorem:

$$\Rightarrow x[\infty] = \lim_{z \rightarrow 1} (1 - z^{-1})X(z) \quad X(z) - (z^{-1})X(z)$$

$$X(z) = x[0] + x[1] z^{-1} + x[2] z^{-2}$$

$$-(z^{-1})X(z) = -x[0] z^{-1} - x[1] z^{-2} - x[2] z^{-3}$$

TABLE 10.1 PROPERTIES OF THE z-TRANSFORM

Section	Property	Signal	z-Transform	ROC
		$x[n]$	$X(z)$	R
		$x_1[n]$	$X_1(z)$	R_1
		$x_2[n]$	$X_2(z)$	R_2
10.5.1	Linearity	$ax_1[n] + bx_2[n]$	$aX_1(z) + bX_2(z)$	At least the intersection of R_1 and R_2
10.5.2	Time shifting	$x[n - n_0]$	$z^{-n_0}X(z)$	R , except for the possible addition or deletion of the origin
10.5.3	Scaling in the z-domain	$e^{j\omega_0 n}x[n]$	$X(e^{-j\omega_0}z)$	R
		$z_0^n x[n]$	$X\left(\frac{z}{z_0}\right)$	$z_0 R$
		$a^n x[n]$	$X(a^{-1}z)$	Scaled version of R (i.e., $ a R$ = the set of points $\{ a z\}$ for z in R)
10.5.4	Time reversal	$x[-n]$	$X(z^{-1})$	Inverted R (i.e., R^{-1} = the set of points z^{-1} , where z is in R)
10.5.5	Time expansion	$x_{(k)}[n] = \begin{cases} x[r], & n = rk \\ 0, & n \neq rk \end{cases}$ for some integer r	$X(z^k)$	$R^{1/k}$ (i.e., the set of points $z^{1/k}$, where z is in R)
10.5.6	Conjugation	$x^*[n]$	$X^*(z^*)$	R
10.5.7	Convolution	$x_1[n] * x_2[n]$	$X_1(z)X_2(z)$	At least the intersection of R_1 and R_2
10.5.7	First difference	$x[n] - x[n - 1]$	$(1 - z^{-1})X(z)$	At least the intersection of R and $ z > 0$
10.5.7	Accumulation	$\sum_{k=-\infty}^n x[k]$	$\frac{1}{1 - z^{-1}}X(z)$	At least the intersection of R and $ z > 1$
10.5.8	Differentiation in the z-domain	$nx[n]$	$-z \frac{dX(z)}{dz}$	R
10.5.9	Initial Value Theorem If $x[n] = 0$ for $n < 0$, then $x[0] = \lim_{z \rightarrow \infty} X(z)$			

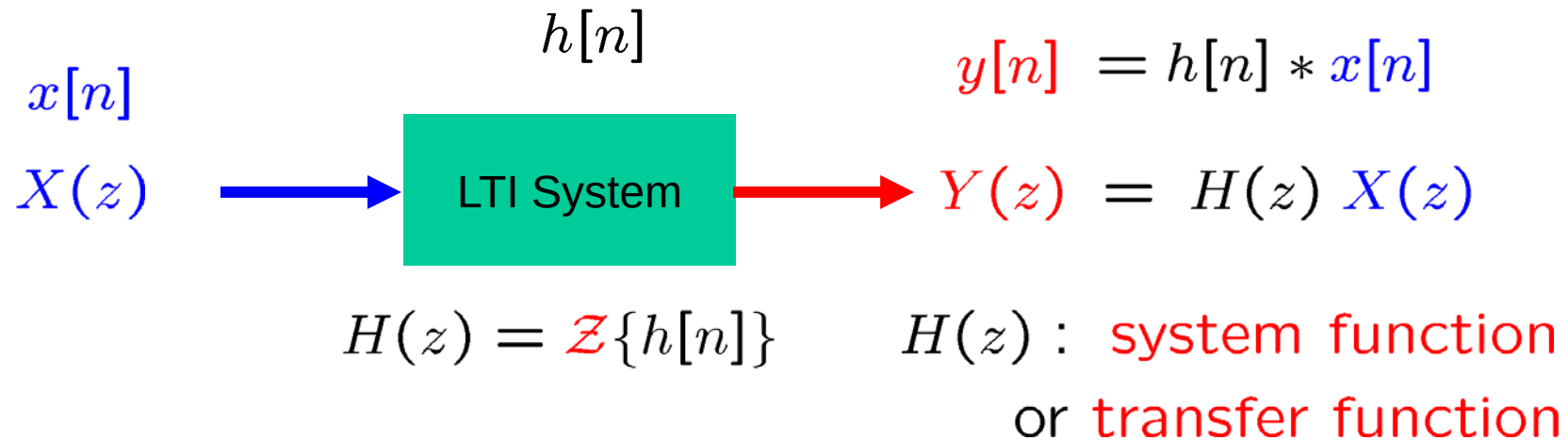
- The z-Transform
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- Geometric Evaluation of the Fourier Transform
- Properties of the z-Transform
- Some Common z-Transform Pairs
- Analysis & Characterization of LTI Systems
Using the z-Transforms
- System Function Algebra and
Block Diagram Representations
- The Unilateral z-Transform

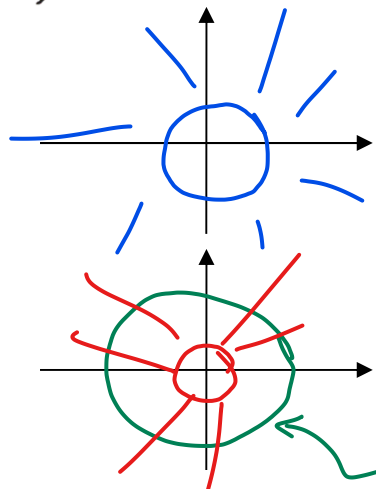
TABLE 10.2 SOME COMMON z-TRANSFORM PAIRS

Signal	Transform	ROC
1. $\delta[n]$	1	All z
2. $u[n]$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
3. $-u[-n - 1]$	$\frac{1}{1 - z^{-1}}$	$ z < 1$
4. $\delta[n - m]$	z^{-m}	All z , except 0 (if $m > 0$) or ∞ (if $m < 0$)
5. $\alpha^n u[n]$	$\frac{1}{1 - \alpha z^{-1}}$	$ z > \alpha $
6. $-\alpha^n u[-n - 1]$	$\frac{1}{1 - \alpha z^{-1}}$	$ z < \alpha $
7. $n\alpha^n u[n]$	$\frac{\alpha z^{-1}}{(1 - \alpha z^{-1})^2}$	$ z > \alpha $
8. $-n\alpha^n u[-n - 1]$	$\frac{\alpha z^{-1}}{(1 - \alpha z^{-1})^2}$	$ z < \alpha $
9. $[\cos \omega_0 n]u[n]$	$\frac{1 - [\cos \omega_0]z^{-1}}{1 - [2 \cos \omega_0]z^{-1} + z^{-2}}$	$ z > 1$
10. $[\sin \omega_0 n]u[n]$	$\frac{[\sin \omega_0]z^{-1}}{1 - [2 \cos \omega_0]z^{-1} + z^{-2}}$	$ z > 1$
11. $[r^n \cos \omega_0 n]u[n]$	$\frac{1 - [r \cos \omega_0]z^{-1}}{1 - [2r \cos \omega_0]z^{-1} + r^2 z^{-2}}$	$ z > r$
12. $[r^n \sin \omega_0 n]u[n]$	$\frac{[r \sin \omega_0]z^{-1}}{1 - [2r \cos \omega_0]z^{-1} + r^2 z^{-2}}$	$ z > r$

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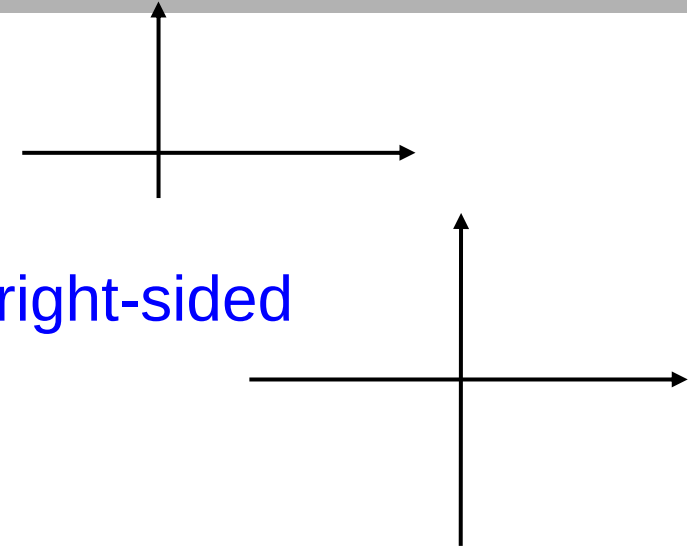
Analysis & Characterization of LTI Systems:



Def.	Conv. (LTI)	Z.T.
Causality $x[m]$ $y[n]$ $m \leq n$ Stability $x[n] < A$ $y[n] < B$ $\forall n$	$h[n]$ $h[n] = 0$ $\forall n < 0$ $\sum_n h[n] < \infty$	$H(z)$  unit circle

■ Causality: $\sum_{n=0}^{+\infty} h[n]z^{-n}$

- For a **causal** LTI system,
 $h[n] = 0$ for $n < 0$, and thus is **right-sided**
- A **DT LTI** system is **causal**
if and only if
the **ROC** of the system function $H(z)$
is the **exterior of a circle** in the z -plane, **including infinity**
- A **DT LTI** system with a **rational** $H(z)$ is **causal**
if and only if
 - (a) **ROC** is exterior of a circle **outside outermost pole**;
and **infinity** must be in the **ROC**; and
 - (b) **order of numerator** $\leq$ **order of denominator**

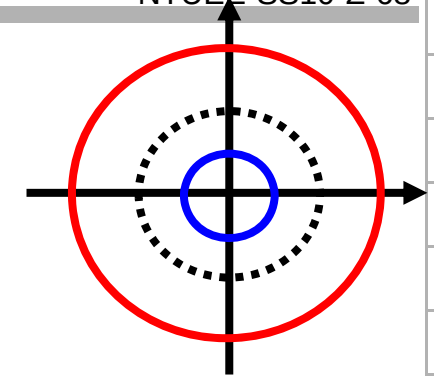


$$H(z) = \frac{b_0 z^M + b_1 z^{M-1} + \dots + b_M}{a_0 z^N + a_1 z^{N-1} + \dots + a_N} = \frac{Y(z)}{X(z)} \quad y[n+N] + \dots = x[n+M] + \dots$$

■ Example 10.21:

$$H(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{1}{1 - 2z^{-1}}, \quad |z| > 2$$

$$H(z) = \frac{2 - \frac{5}{2}z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})} = \frac{2z(z - \frac{5}{4})}{(z - \frac{1}{2})(z - 2)} = \frac{2z^2 - \frac{5}{2}z}{z^2 - \frac{5}{2}z + 1}$$



⇒ *ROC*: the exterior of a circle of outside the outermost pole

⇒ the impulse response is right-sided

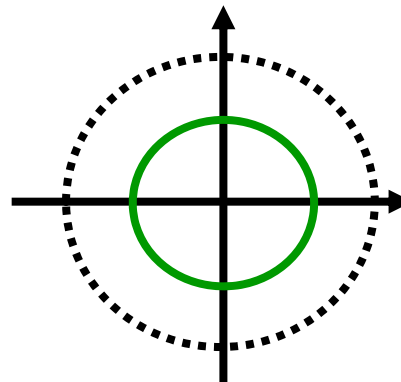
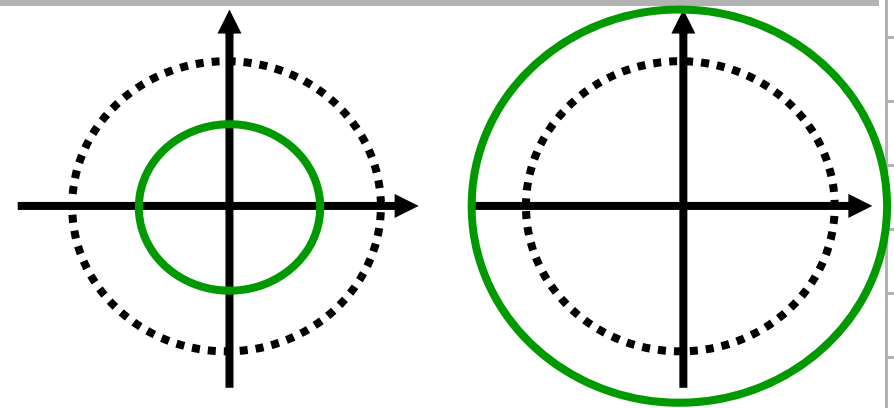
⇒ deg of num of $H(z)$ = deg of den of $H(z)$

⇒ the system is causal

$$\Rightarrow h[n] = \left[\left(\frac{1}{2} \right)^n + 2^n \right] u[n] \quad \Rightarrow h[n] = 0, n < 0$$

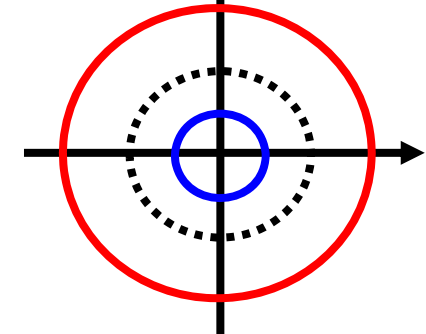
■ Stability:

- An **DT LTI** system is **stable** if and only if the **ROC** of **$H(z)$** includes the **unit circle** [i.e., $|z| = 1$]
- A **causal** LTI system with **rational $H(z)$** is **stable** if and only if **all of the poles of $H(z)$** lie in the **inside the unit circle**, i.e., all of the poles have **magnitude < 1**



■ Example 10.22:

$$H(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{1}{1 - 2z^{-1}}, \quad |z| > 2$$



$\Rightarrow$ *ROC* does not include the unit circle $\Rightarrow$ **NOT stable**

$$\Rightarrow \text{i.e., } h[n] = \left[\left(\frac{1}{2} \right)^n + 2^n \right] u[n] \rightarrow \infty, \text{ as } n \rightarrow \infty$$

• If *ROC* = $|z| < 1/2$ $\Rightarrow h[n] = - \left[\left(\frac{1}{2} \right)^n + 2^n \right] u[-n - 1]$

$\Rightarrow$ the system is **neither causal nor stable**

• If *ROC* = $1/2 < |z| < 2$ $\Rightarrow h[n] = \left(\frac{1}{2} \right)^n u[n] - 2^n u[-n - 1]$

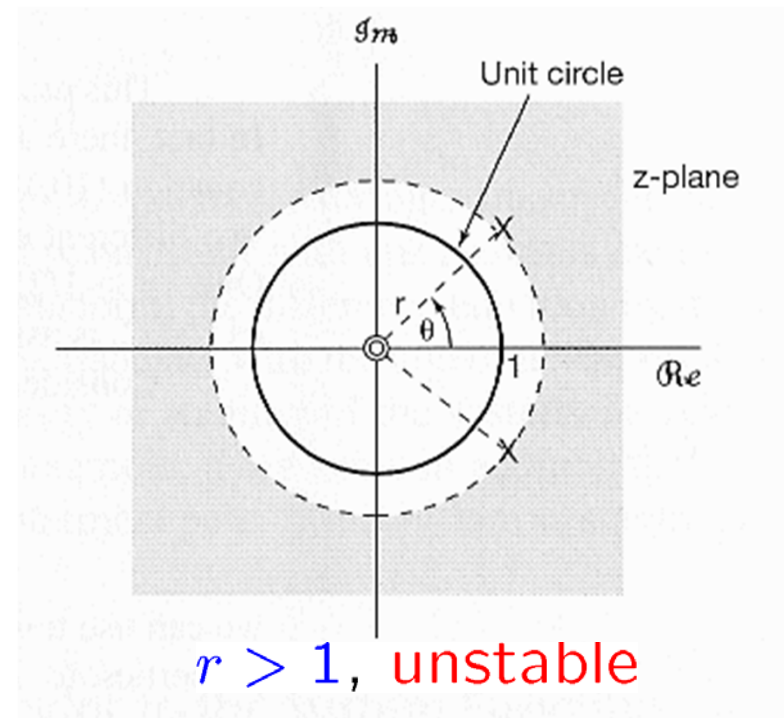
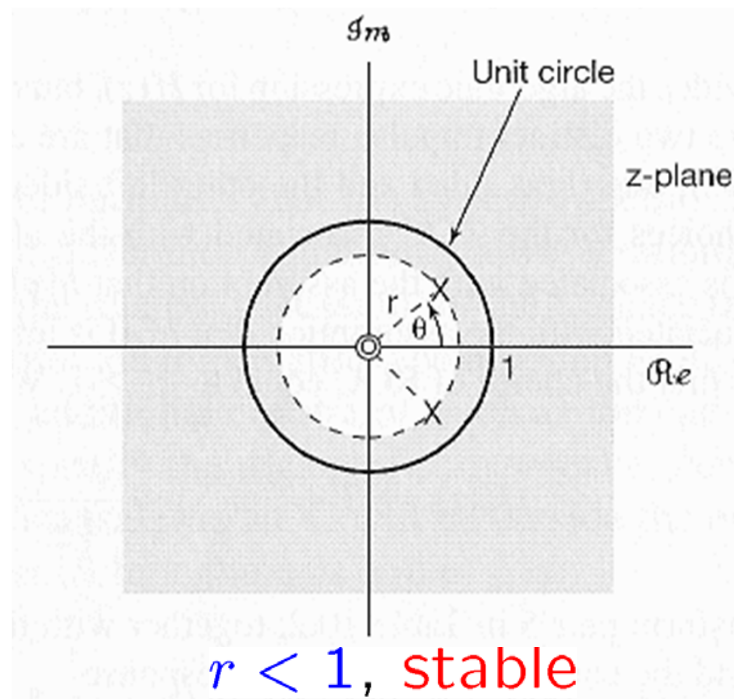
$\Rightarrow$ the system is **NOT causal**, but **stable**

■ Example 10.24:

$$H(z) = \frac{1}{1 - (2r \cos \theta)z^{-1} + r^2 z^{-2}} = \frac{z^2}{z^2 - (2r \cos \theta)z + r^2}$$

$$\Rightarrow z_1 = re^{j\theta}, \quad z_2 = re^{-j\theta}$$

If it is **causal**, $|z| > |r|$

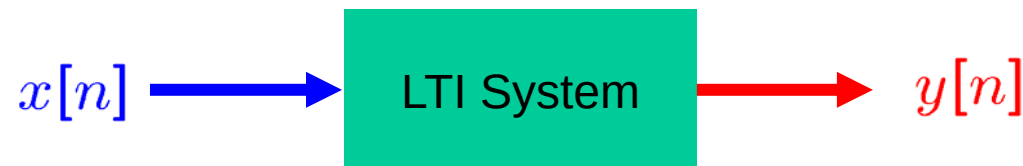


- LTI Systems by Linear Constant-Coef Difference Equations:

$$a_0y[n] + a_1y[n-1] + \cdots + a_{N-1}y[n-N+1] + a_Ny[n-N]$$

$$= b_0x[n] + b_1x[n-1] + \cdots + b_{M-1}x[n-M+1] + b_Mx[n-M]$$

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k]$$



$$Y(z) = X(z)H(z) \qquad H(z) = \frac{Y(z)}{X(z)}$$

$$x[n - n_0] \xleftrightarrow{Z} z^{-n_0} X(z)$$

$$\mathcal{Z} \left\{ \sum_{k=0}^N a_k y[n - k] \right\} = \mathcal{Z} \left\{ \sum_{k=0}^M b_k x[n - k] \right\}$$

$$\sum_{k=0}^N a_k \mathcal{Z} \left\{ y[n - k] \right\} = \sum_{k=0}^M b_k \mathcal{Z} \left\{ x[n - k] \right\}$$

$$\sum_{k=0}^N a_k z^{-k} Y(z) = \sum_{k=0}^M b_k z^{-k} X(z)$$

$$\Rightarrow H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}} = \frac{b_0 + b_1 z^{-1} + \dots + b_M z^{-M}}{a_0 + a_1 z^{-1} + \dots + a_N z^{-N}}$$

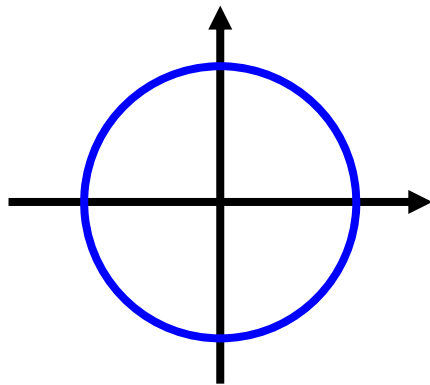
$$= \frac{b_0 z^N + b_1 z^{N-1} + \dots + b_M z^{N-M}}{a_0 z^N + a_1 z^{N-1} + \dots + a_N}$$

zeros
poles

■ Example 10.25:

$$y[n] - \frac{1}{2}y[n-1] = x[n] + \frac{1}{3}x[n-1]$$

$$\Rightarrow Y(z) - \frac{1}{2}z^{-1}Y(z) = X(z) + \frac{1}{3}z^{-1}X(z)$$



$$\Rightarrow H(z) = \frac{1 + \frac{1}{3}z^{-1}}{1 - \frac{1}{2}z^{-1}} = \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{\frac{1}{3}z^{-1}}{1 - \frac{1}{2}z^{-1}}$$

$$= \frac{z + \frac{1}{3}}{z - \frac{1}{2}} = \frac{z}{z - \frac{1}{2}} + \frac{\frac{1}{3}}{z - \frac{1}{2}}$$

- If $ROC = \{|z| > 1/2\}$, $\Rightarrow h[n]$ is right-sided

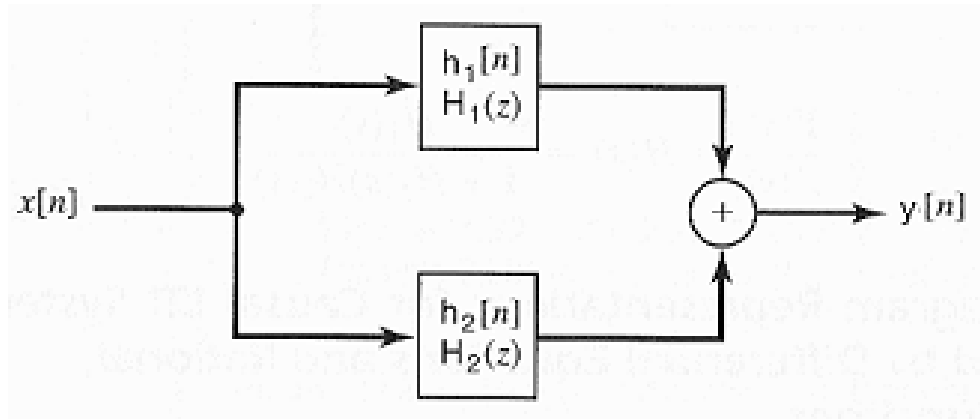
$$\Rightarrow h[n] = \left(\frac{1}{2}\right)^n u[n] + \frac{1}{3}\left(\frac{1}{2}\right)^{n-1} u[n-1]$$

- If $ROC = \{|z| < 1/2\}$, $\Rightarrow h[n]$ is left-sided

$$\Rightarrow h[n] = -\left(\frac{1}{2}\right)^n u[-n-1] - \frac{1}{3}\left(\frac{1}{2}\right)^{n-1} u[-n]$$

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System Function Blocks:



- parallel interconnection

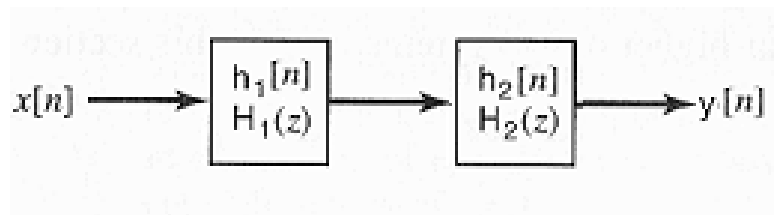
$$h[n] = h_1[n] + h_2[n]$$

$$H(z) = H_1(z) + H_2(z)$$

- series interconnection

$$h[n] = h_1[n] * h_2[n]$$

$$H(z) = H_1(z) H_2(z)$$



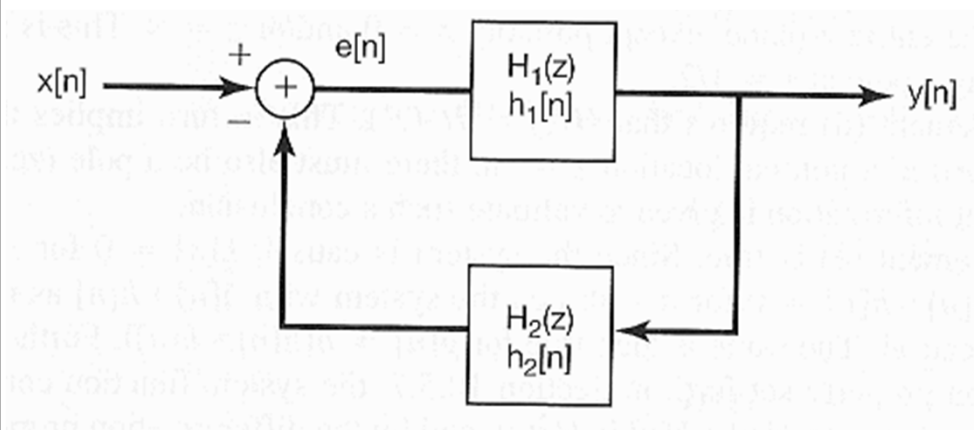
- feedback interconnection

$$Y = H_1 E$$

$$Z = H_2 Y$$

$$E = X - Z$$

$$H(z) = \frac{H_1(z)}{1 + H_1(z)H_2(z)}$$



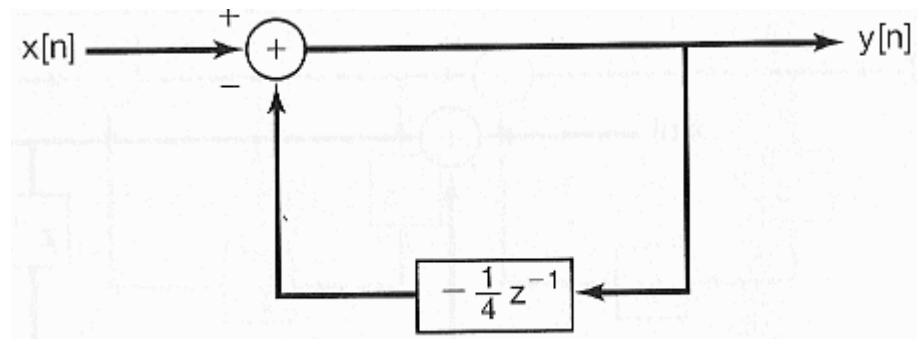
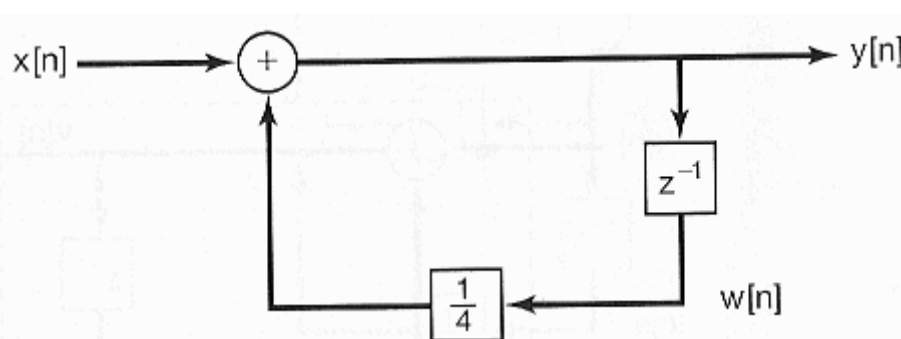
■ Example 10.28:

- Consider a **causal LTI** system with system function

$$H(z) = \frac{1}{1 - \frac{1}{4}z^{-1}}$$

$$\Rightarrow y[n] = \frac{1}{1 - \frac{1}{4}z^{-1}}x[n] \Rightarrow y[n] - \frac{1}{4}y[n-1] = x[n]$$

$$\Rightarrow y[n] = \frac{1}{4}y[n-1] + x[n]$$

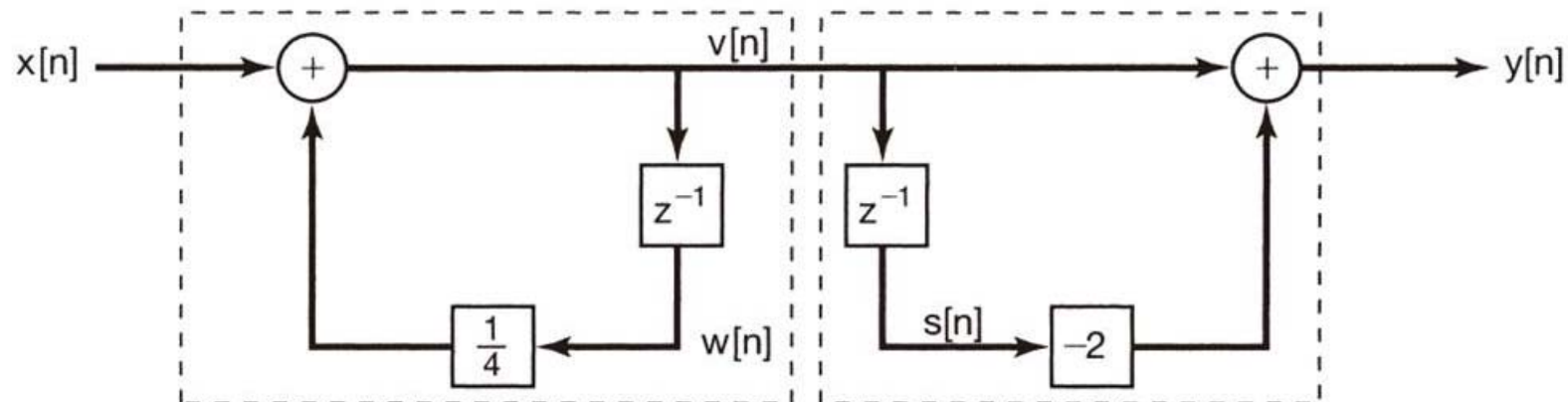


$$w[n] = y[n-1]$$

■ Example 10.29:

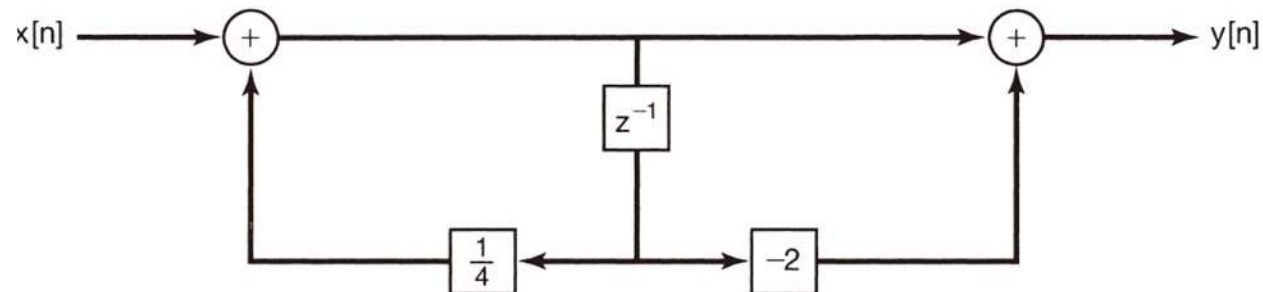
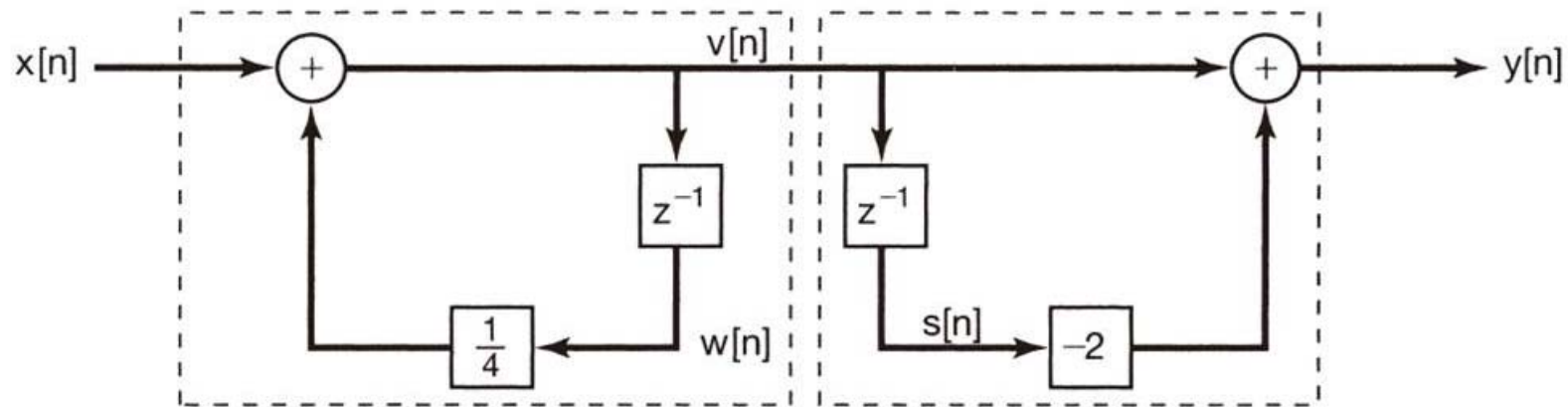
$$\frac{Y(z)}{X(z)} = H(z) = \frac{1 - 2z^{-1}}{1 - \frac{1}{4}z^{-1}} = \left(\frac{1}{1 - \frac{1}{4}z^{-1}} \right) (1 - 2z^{-1})$$

$$Y(z) = (1 - 2z^{-1})V(z) \quad V(z) = \left(\frac{1}{1 - \frac{1}{4}z^{-1}} \right) X(z)$$



■ Example 10.29:

$$H(z) = \frac{1 - 2z^{-1}}{1 - \frac{1}{4}z^{-1}} = \left(\frac{1}{1 - \frac{1}{4}z^{-1}} \right) (1 - 2z^{-1})$$

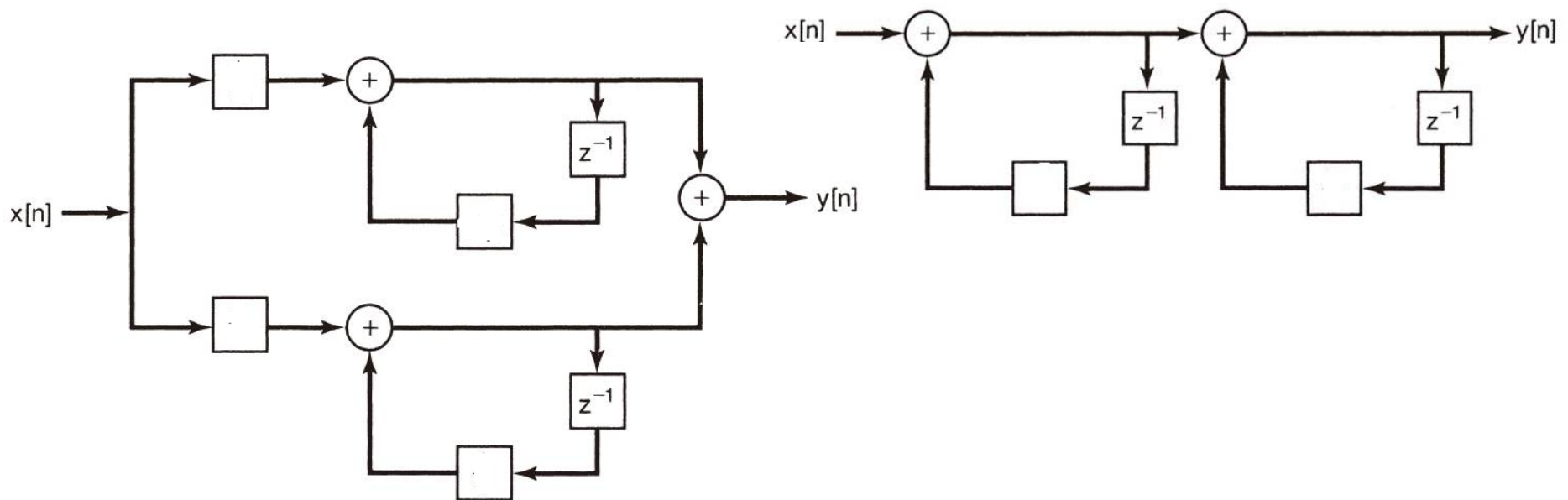
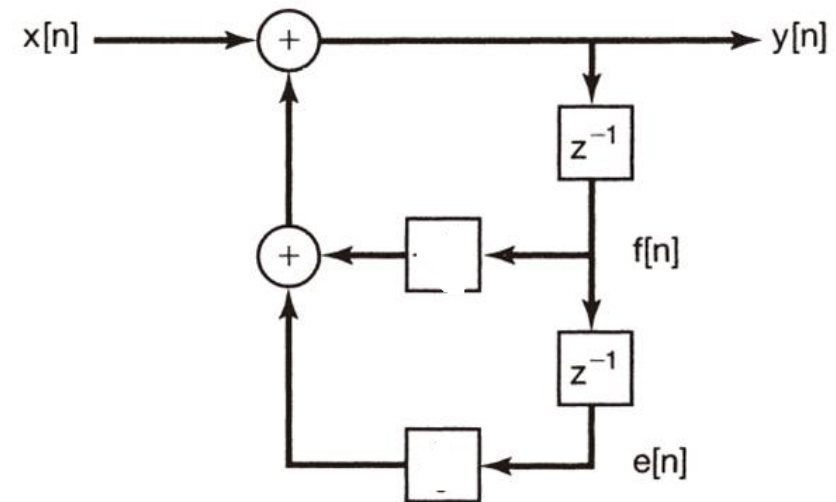


■ Example 10.30:

$$H(z) = \frac{1}{1 + \frac{1}{4}z^{-1} - \frac{1}{8}z^{-2}}$$

$$H(z) = \left(\frac{1}{1 + \frac{1}{2}z^{-1}} \right) \left(\frac{1}{1 - \frac{1}{4}z^{-1}} \right)$$

$$H(z) = \frac{\frac{2}{3}}{(1 + \frac{1}{2}z^{-1})} + \frac{\frac{1}{3}}{(1 - \frac{1}{4}z^{-1})}$$



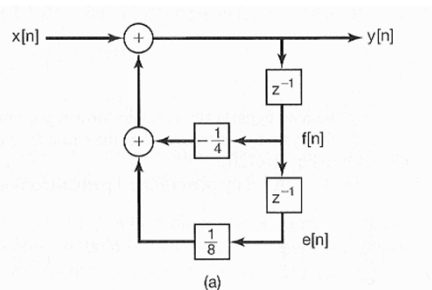
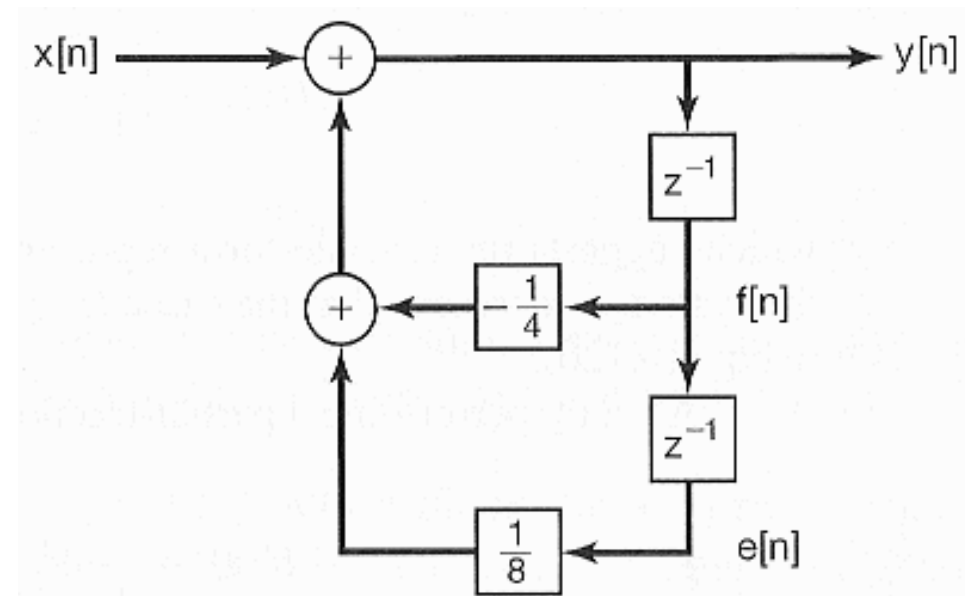
■ Example 10.30:

$$H(z) = \frac{1}{(1 + \frac{1}{2}z^{-1})(1 - \frac{1}{4}z^{-1})} = \frac{1}{1 + \frac{1}{4}z^{-1} - \frac{1}{8}z^{-2}}$$

$$\Rightarrow y[n] + \frac{1}{4}y[n-1] - \frac{1}{8}y[n-2] = x[n]$$

$$\Rightarrow y[n] = -\frac{1}{4}y[n-1] + \frac{1}{8}y[n-2] + x[n]$$

$$\Rightarrow \begin{cases} y[n-1] = f[n] \\ y[n-2] = e[n] = f[n-1] \end{cases}$$

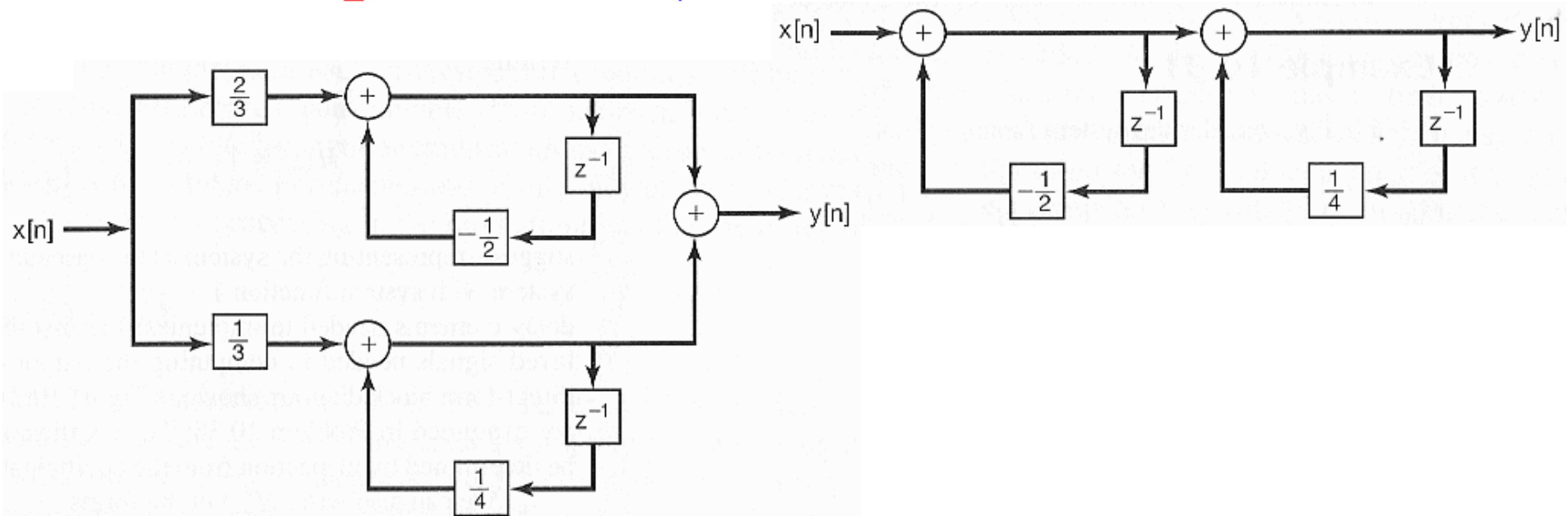
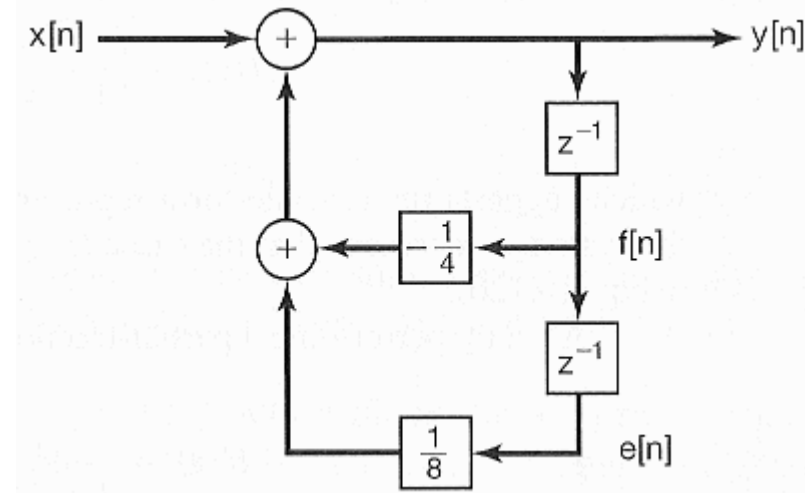


■ Example 10.30:

$$H(z) = \frac{1}{1 + \frac{1}{4}z^{-1} - \frac{1}{8}z^{-2}}$$

$$H(z) = \left(\frac{1}{1 + \frac{1}{2}z^{-1}} \right) \left(\frac{1}{1 - \frac{1}{4}z^{-1}} \right)$$

$$H(z) = \frac{\frac{2}{3}}{(1 + \frac{1}{2}z^{-1})} + \frac{\frac{1}{3}}{(1 - \frac{1}{4}z^{-1})}$$



- The z-Transform
- The Region of Convergence for z-Transforms
- The Inverse z-Transform
- Geometric Evaluation of the Fourier Transform
- Properties of the z-Transform
- Some Common z-Transform Pairs
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Using the z-Transforms
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Block Diagram Representations
- The Unilateral z-Transform

■ The Unilateral z-Transform of x(t):

bilateral zT

$$X(z) \triangleq \sum_{n=-\infty}^{+\infty} x[n]z^{-n}$$

$$= \sum_{n=-\infty}^{-1} x[n]z^{-n} + \sum_{n=0}^{+\infty} x[n]z^{-n}$$

unilateral zT

for causal system &
with nonzero init. cond.

$$\mathcal{X}(z) \triangleq \sum_{n=0}^{+\infty} x[n]z^{-n}$$

$$x[n] \xleftrightarrow{\mathcal{Z}} X(z)$$

$$X(z) = \mathcal{Z}\{x[n]\}$$

$$x[n] = \mathcal{Z}^{-1}\{X(z)\}$$

$$x[n] \xleftrightarrow{\mathcal{U}\mathcal{Z}} \mathcal{X}(z)$$

$$\mathcal{X}(z) = \mathcal{U}\mathcal{Z}\{x[n]\}$$

$$x[n] = \mathcal{U}\mathcal{Z}^{-1}\{\mathcal{X}(z)\}$$

ROC : exterior of a circle

■ Time-Shifting Property

$$x[n] \xleftrightarrow{\mathcal{U}\mathcal{Z}} \mathcal{X}(z)$$

$$\mathcal{X}(z) = \sum_{n=0}^{+\infty} x[n]z^{-n} = x[0] + x[1]z^{-1} + x[2]z^{-2} + \dots$$

$$x[n-1] \xleftrightarrow{\mathcal{U}\mathcal{Z}} z^{-1} \mathcal{X}(z) + x[-1]$$

$$\sum_{n=0}^{+\infty} x[n-1]z^{-n} = x[-1] + x[0]z^{-1} + x[1]z^{-2} + x[2]z^{-3} + \dots$$

$$z^{-1} \mathcal{X}(z) = x[0]z^{-1} + x[1]z^{-2} + x[2]z^{-3} + \dots$$

$$x[n-2] \xleftrightarrow{\mathcal{U}\mathcal{Z}} z^{-2} \mathcal{X}(z) + x[-1]z^{-1} + x[-2]$$

$$x[n+1] \xleftrightarrow{\mathcal{U}\mathcal{Z}} z \mathcal{X}(z) - zx[0]$$

■ Example 10.33:

$$x[n] = a^{n+1}u[n+1]$$

- since $x[-1] = 1 \neq 0$, $X(z) \neq \mathcal{X}(z)$

$\Rightarrow$ bilateral transform :
$$X(z) = \frac{z}{1 - az^{-1}}, \quad |z| > |a|$$

$\Rightarrow$ unilateral transform :
$$\begin{aligned}\mathcal{X}(z) &= \sum_{n=0}^{+\infty} x[n]z^{-n} \\ &= \sum_{n=0}^{+\infty} a^{n+1}z^{-n} \\ &= \frac{a}{1 - az^{-1}}, \quad |z| > |a|\end{aligned}$$

■ Example 10.33:

$$x_1[n] = a^n u[n] \quad \mathcal{X}_1(z) = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}, \quad |z| > |a|$$

$$x_2[n] = x_1[n + 1] \quad \mathcal{X}_2(z) = z \frac{z}{z - a} - z x_1[0]$$

$$= a^{n+1} u[n + 1] \quad = \frac{z^2}{z - a} - z \cdot 1$$

$$= \frac{z^2 - z^2 + az}{z - a}$$

$$= \frac{az}{z - a}$$

$$= \frac{a}{1 - z^{-1}}$$

TABLE 10.3 PROPERTIES OF THE UNILATERAL z-TRANSFORM

Property	Signal	Unilateral z-Transform
—	$x[n]$	$\mathcal{X}(z)$
—	$x_1[n]$	$\mathcal{X}_1(z)$
—	$x_2[n]$	$\mathcal{X}_2(z)$
Linearity	$ax_1[n] + bx_2[n]$	$a\mathcal{X}_1(z) + b\mathcal{X}_2(z)$
Time delay	$x[n-1]$	$z^{-1}\mathcal{X}(z) + x[-1]$
Time advance	$x[n+1]$	$z\mathcal{X}(z) - zx[0]$
Scaling in the z-domain	$e^{j\omega_0 n} x[n]$	$\mathcal{X}(e^{-j\omega_0} z)$
	$z_0^n x[n]$	$\mathcal{X}(z/z_0)$
	$a^n x[n]$	$\mathcal{X}(a^{-1} z)$
Time expansion	$x_k[n] = \begin{cases} x[m], & n = mk \\ 0, & n \neq mk \end{cases} \text{ for any } m$	$\mathcal{X}(z^k)$
Conjugation	$x^*[n]$	$\mathcal{X}^*(z^*)$
Convolution (assuming that $x_1[n]$ and $x_2[n]$ are identically zero for $n < 0$)	$x_1[n] * x_2[n]$	$\mathcal{X}_1(z)\mathcal{X}_2(z)$
First difference	$x[n] - x[n-1]$	$(1 - z^{-1})\mathcal{X}(z) - x[-1]$
Accumulation	$\sum_{k=0}^n x[k]$	$\frac{1}{1 - z^{-1}} \mathcal{X}(z)$
Differentiation in the z-domain	$nx[n]$	$-z \frac{d\mathcal{X}(z)}{dz}$
Initial Value Theorem		
$x[0] = \lim_{z \rightarrow \infty} \mathcal{X}(z)$		

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{jw}) e^{jwn} dw$$

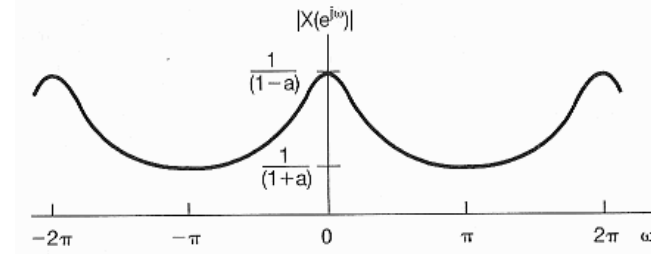
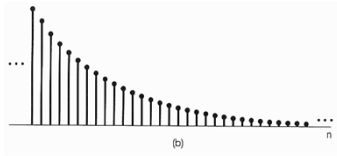
$$X(e^{jw}) = \sum_{n=-\infty}^{+\infty} x[n] e^{-jwn}$$

$$x[n] = \frac{1}{2\pi j} \oint X(z) z^{n-1} dz$$

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n] z^{-n}$$

$$X(z) = X(re^{jw}) = \mathcal{Z} \{x[n]\} = \mathcal{F} \{x[n] r^{-n}\}$$

$$X(e^{jw}) = \mathcal{F} \{x[n]\} = \mathcal{Z} \{x[n]\} \Big|_{z=e^{jw}} = X(z) \Big|_{z=e^{jw}}$$



$$x[n] = a^n u[n], \quad |a| < 1 \quad \xleftrightarrow{\mathcal{F}} \quad X(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}}$$

$$x[n] = a^n u[n] \quad \xleftrightarrow{\mathcal{L}} \quad X(z) = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}$$

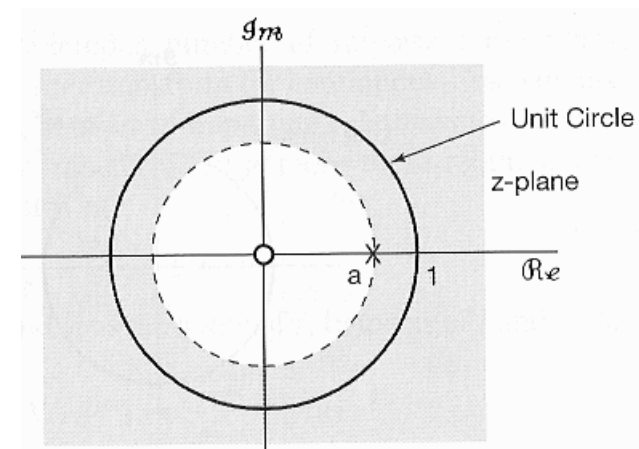
$$|z| > |a|$$

definition
theorem
property

Causality

Stability

ROC



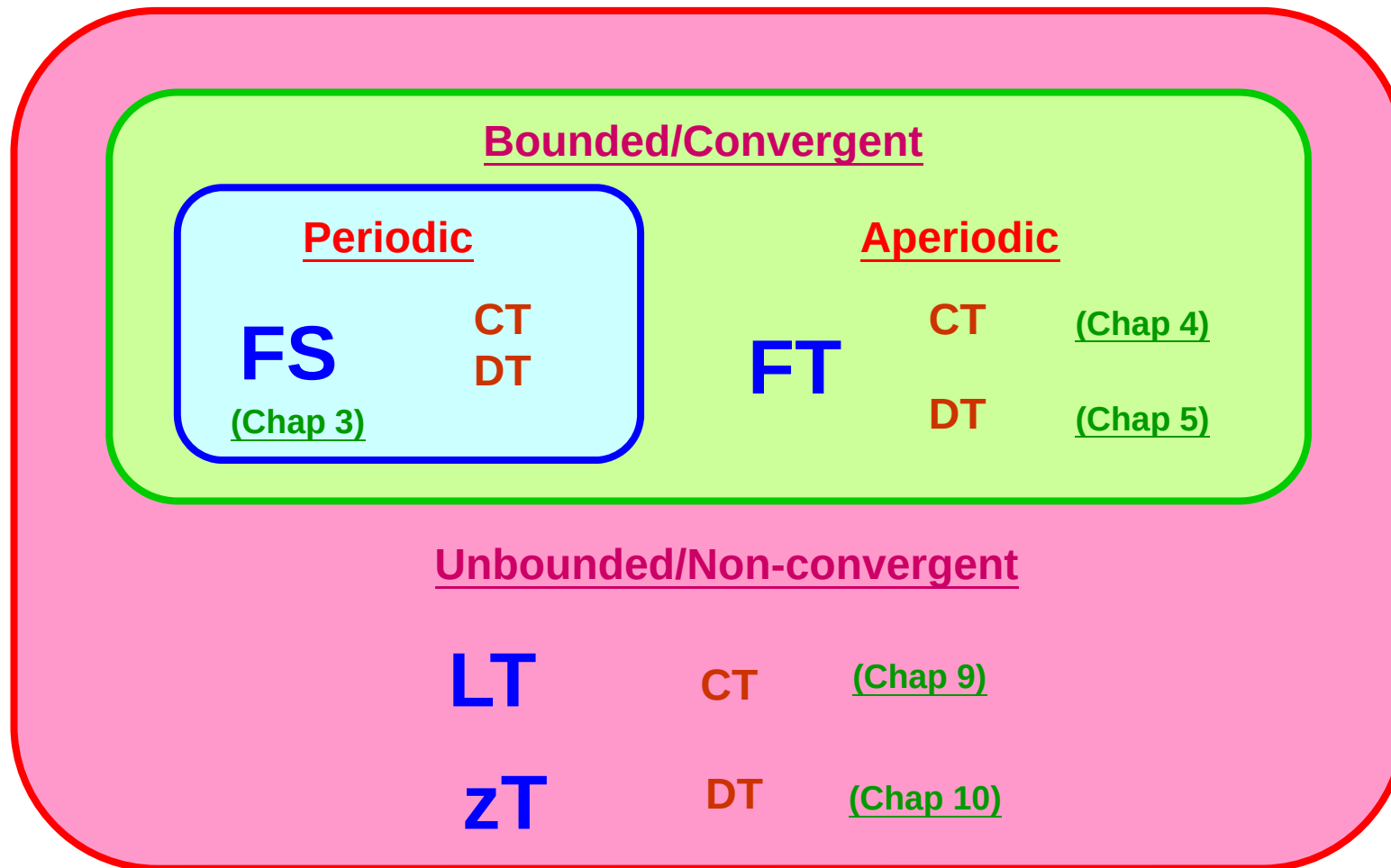
- The z-Transform
- The ROC for z-T
- The Inverse z-T
- Geometric Evaluation of the FT
- Properties of the z-T
 - Linearity
 - Time Reversal
 - Convolution
 - Differentiation in the z-Domain
 - Time Shifting
 - Time Expansion
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Introduction

(Chap 1)

LTI & Convolution

(Chap 2)



Time-Frequency (Chap 6)

Communication (Chap 8)

Digital
Signal
Processing (dsp-8)

CT-DT (Chap 7)

Control (Chap 11)